

Bias and Fairness in LLMs

Trustworthy Large Language Models

ELL881 · AIL821



Yatin Nandwani
Research Scientist, IBM

Bias and Fairness in LLMs



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Business News > Technology > Artificial Intelligence > Meta sued by 26 ex-employees over alleged AI-driven layoff bias

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The lawsuit, filed in the US District Court in Oakland, California, claims Meta relied on AI-assisted performance metrics —including productivity levels and AI token usage — when selecting employees for job cuts.



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The lawsuit, filed in the US District Court in Oakland, California, alleges that Meta used AI-driven metrics—including productivity levels and AI token usage—to

hai.stanford.edu/news/ai-hiring-tools-can-yield-racial-bias-and-systemic-rejection

NEWS

AI Hiring Tools Can Yield Racial Bias and Systemic Rejection



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World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ Technology ▾ Investig

Workday must face California lawsuit over AI bias in job screening tools

By Daniel Wiessner

June 23, 2026 1:42 AM GMT+5:30 · Updated June 23, 2026



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Bias and F

MEDIA

AI chatbots take heat over left-wing bias: 'No longer be considered neutral'

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Bias and Fairness in LLMs

What is wrong here?

Who is harmed — and how?

Is it bias? Is it unfairness? Are those the same thing?



Bias and Fairness in LLMs

What is
Fairness/Bias?



Bias and Fairness in LLMs

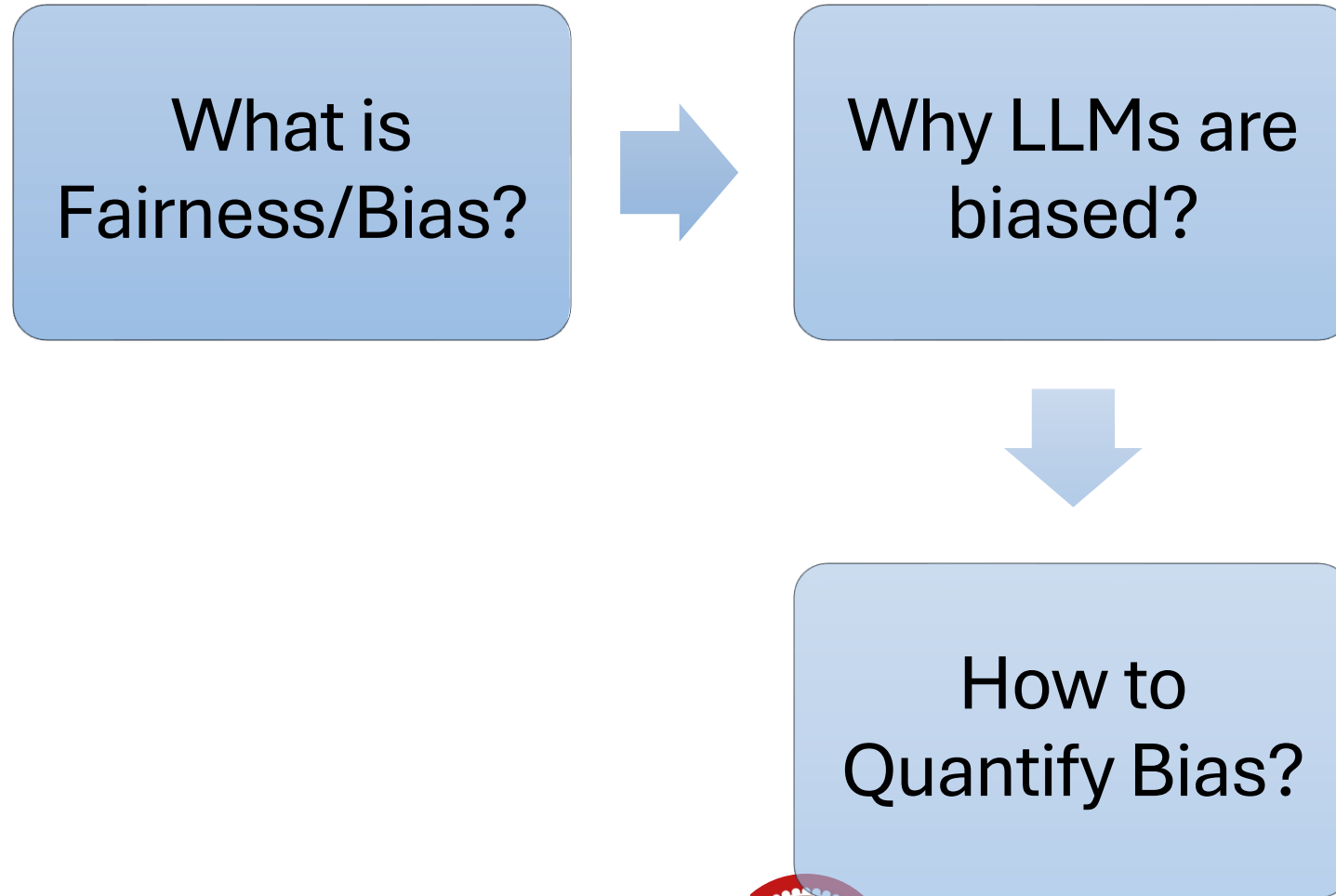
What is
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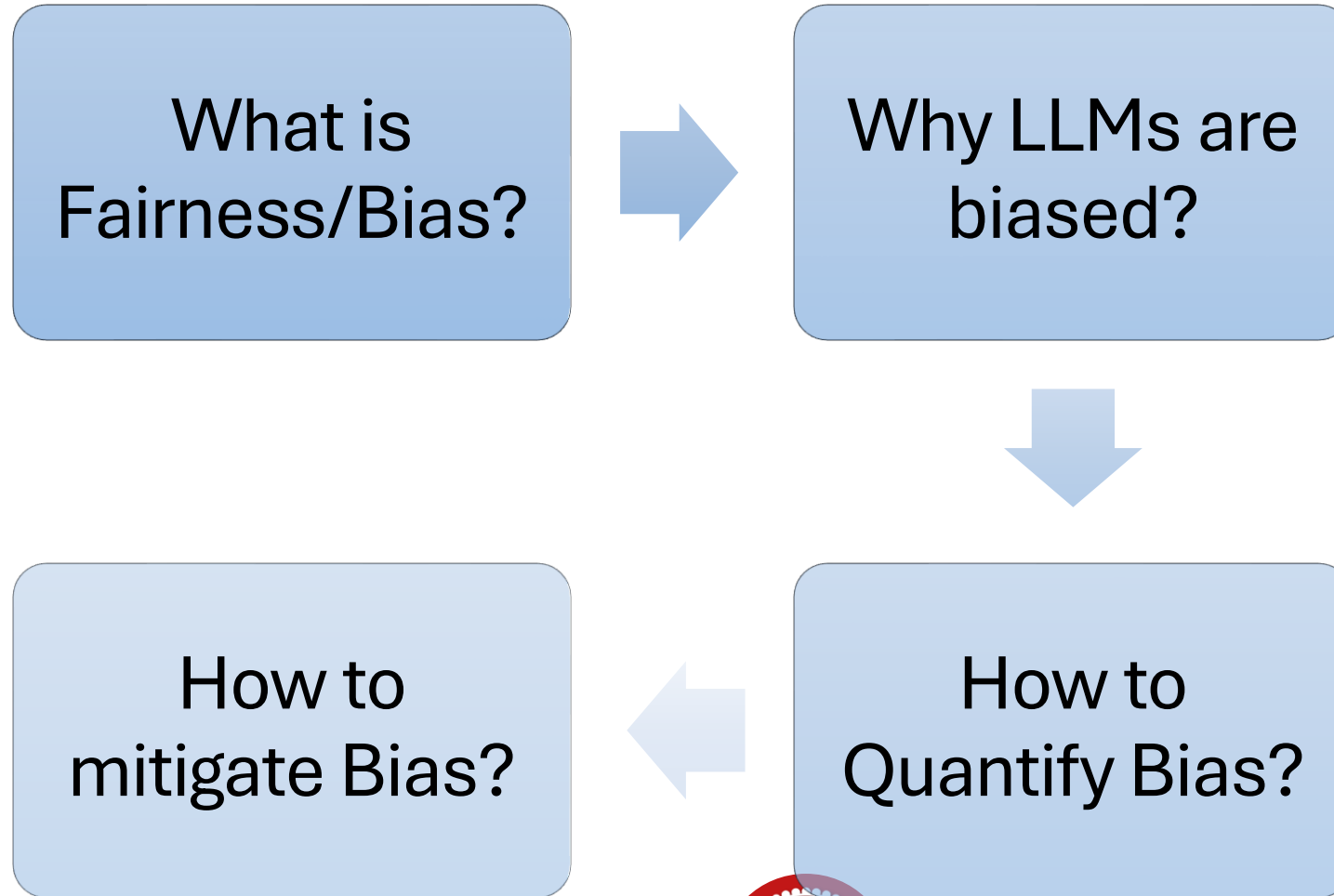
Why LLMs are
biased?



Bias and Fairness in LLMs



Bias and Fairness in LLMs



Bias and Fairness in LLMs

What is
Fairness/Bias?



One Word, Four Meanings

You already use “bias” in three other senses. None of them is the one in the headlines.

Statistics

Machine learning

AI in general – Inductive bias

Social science

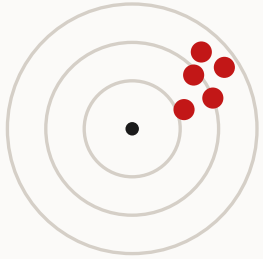
Caliskan, Bryson & Narayanan, Science (2017), citing Bishop (2006) for the third sense.



One Word, Four Meanings

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$$\text{bias}(\theta) = E[\theta] - \theta$$

The estimator is systematically off-centre. A property of a procedure against a ground truth.

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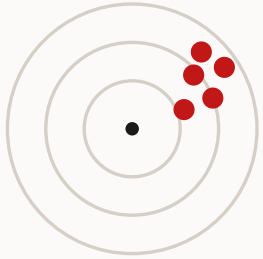
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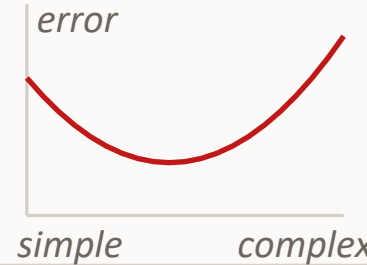
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Machine learning



$$\text{error} = \text{bias}^2 + \text{variance} + \sigma^2$$

Too simple a model under-fits.
Nothing to do with people.

AI in general – Inductive bias

Social science

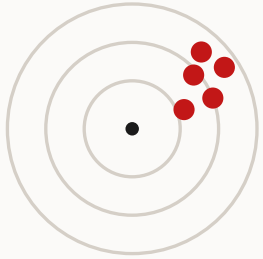
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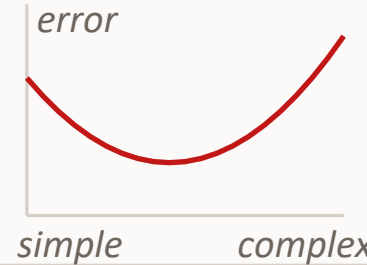
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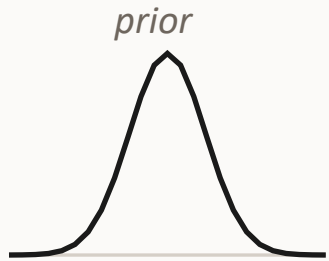
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bias = prior information

“A necessary prerequisite for intelligence.” No prior, no generalisation.

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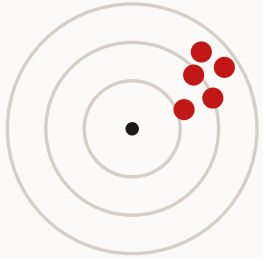
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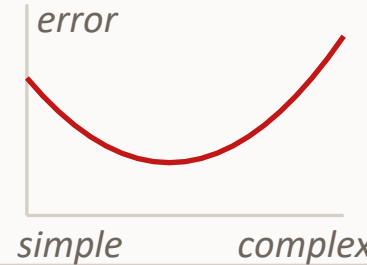
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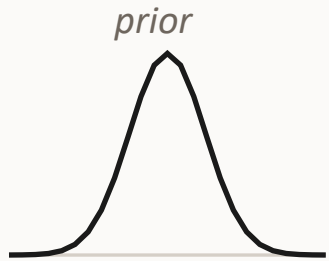
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A prior learned from human culture can carry its harms. That is the fourth sense.

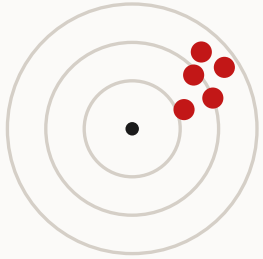
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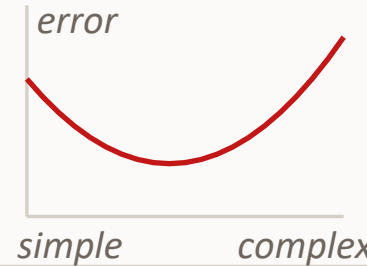
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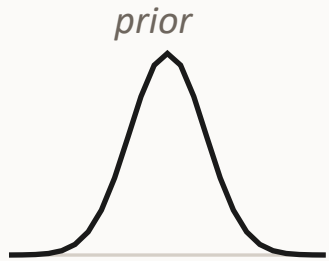
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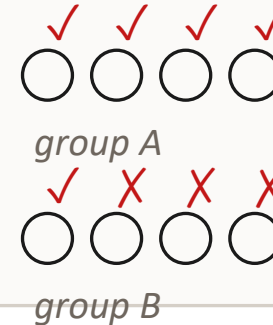
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social bias

A systematic skew against a social group. The only one of the four with a victim.

A prior learned from human culture can carry its harms. That is the fourth sense.

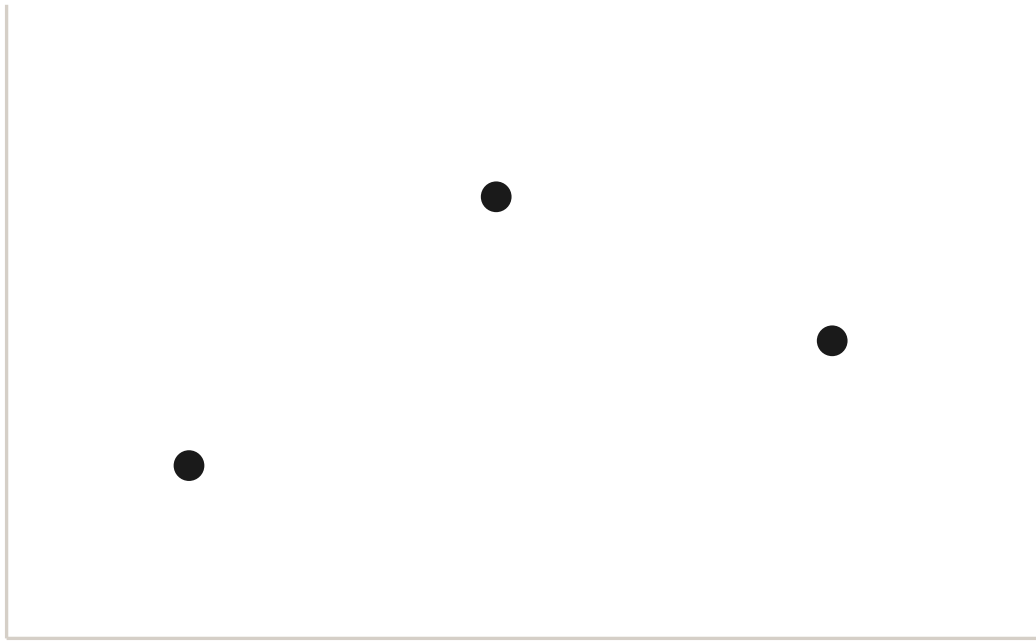
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To Be Intelligent Is to Be Biased

The third sense, taken seriously — because it is the one that makes learning possible.

Which curve?



three observations

Caliskan, Bryson & Narayanan, Science (2017), citing Bishop (2006). Curves are illustrative.

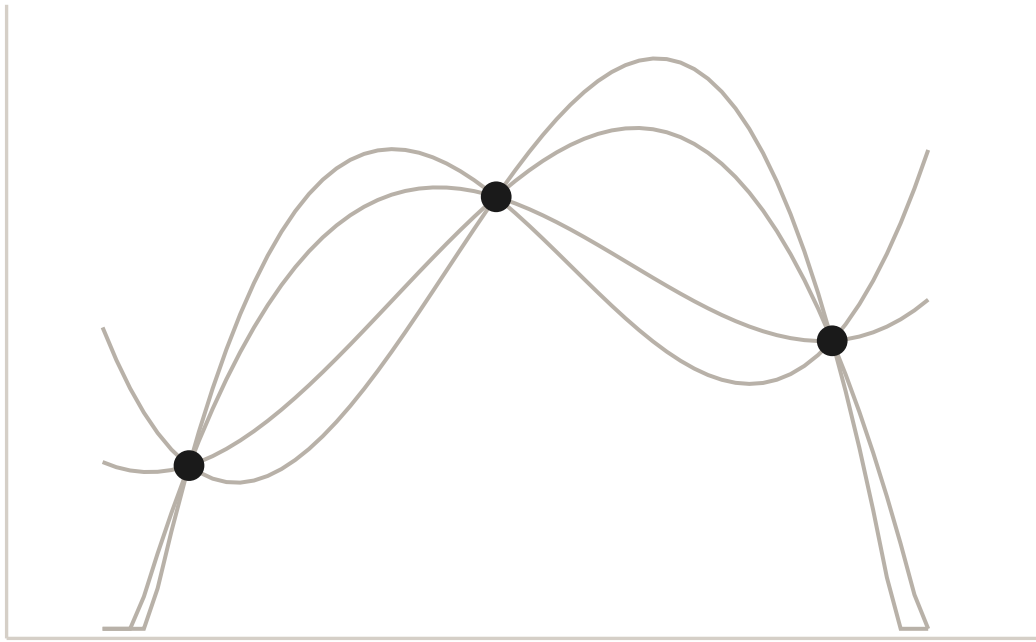


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Which curve?

Infinitely many pass through the same three points.
The data alone does not choose one.



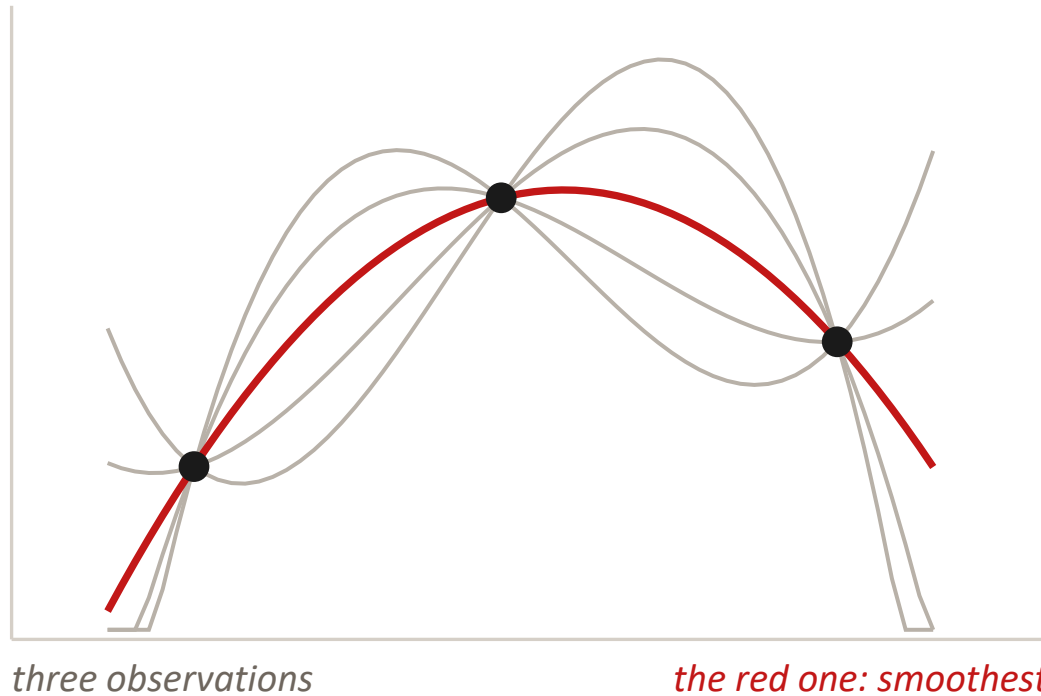
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The choice is the prior.

“Prefer the smooth one” is an assumption you brought, not a fact you measured.



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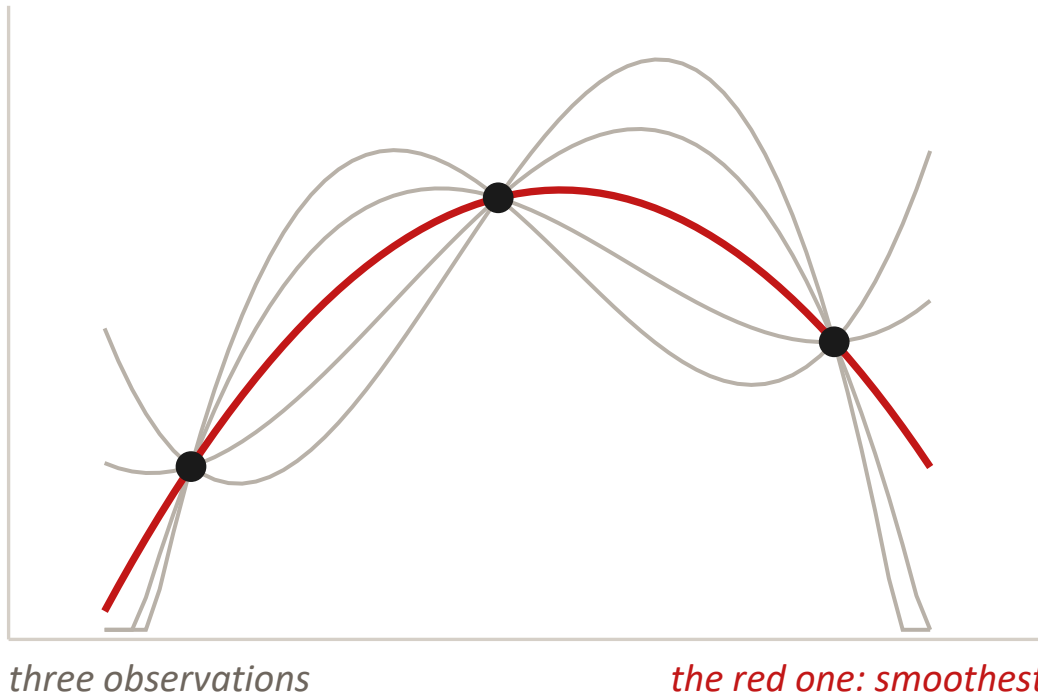
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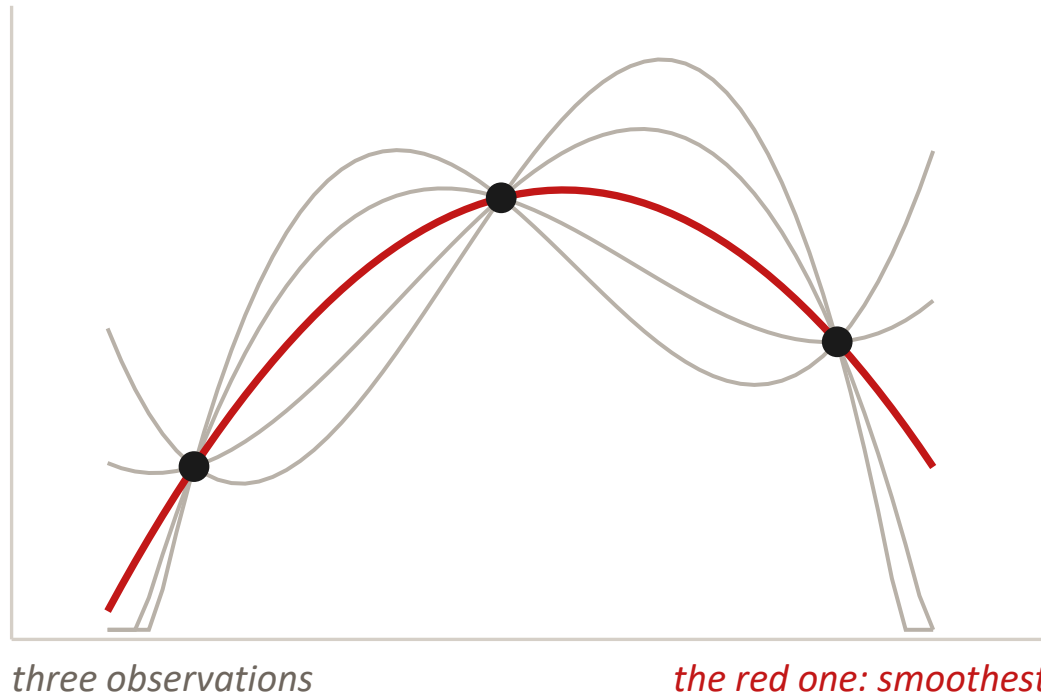
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No prior, no learning. The question is which prior.



To Be Biased Is Not Always to Be Intelligent

Where does the prior come from?

The doctor said that ____ would be late.

The nurse said that ____ would be late.

Caliskan, Bryson & Narayanan, Science (2017). Completions illustrative; the measurement comes later.



To Be Biased Is Not Always to Be Intelligent

Where does the prior come from?

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The prior was learned from text people wrote. The precedent it encodes — who does which job — is one we have recognised as harmful.

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“...bias can be problematic where prior information is derived from precedents known to be harmful.”

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“...bias can be problematic where prior information is derived from precedents known to be harmful.”

Such a prior is not intelligence. That is the bias we mean today.

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Bias in the Context of Today's Lecture

“an output that is not only **skewed**, but skewed in a way which is **unfair**”

Centre for Data Ethics and Innovation, Review into Bias in Algorithmic Decision-Making (2020)



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skewed

A systematic tilt. A measurable fact about the output — you can count it.



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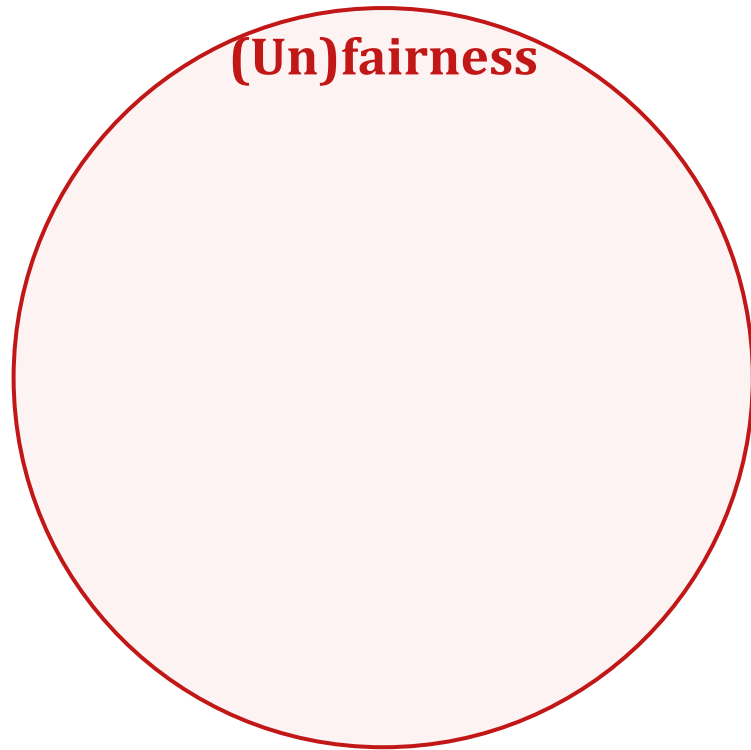
A judgement, not a count. It needs a standard of fairness to be measured against.

Skew alone is not bias. Bias presupposes a definition of (un)fairness.



Unfairness, Bias, Discrimination

How the three words nest?



After CDEI (2020), Review into Bias in Algorithmic Decision-Making — redrawn.



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Unfairness, Bias, Discrimination

How the three words nest?

Synonyms

equity. (FAIRNESS)

impartiality.

justice (FAIRNESS)

fairness *noun* [U] (FAIR TREATMENT)

C1

the quality of treating people equally or in a way that is right or reasonable:

- *He had a real sense of fairness and hated injustice.*
- *The ban on media reporting has made some people question the fairness of the election (= ask whether it was fair).*

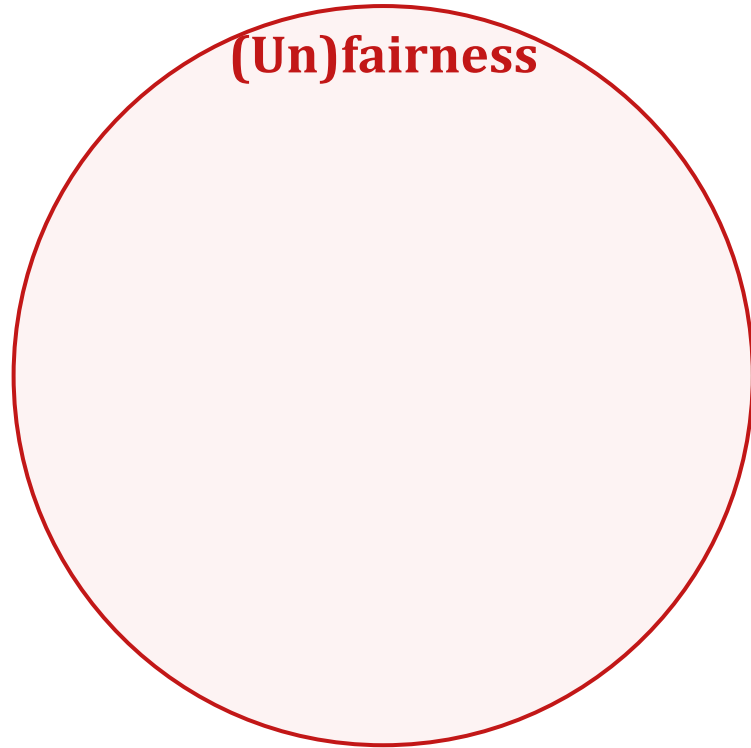
(Un)fairness

After CDEI (2020), Review into Bias in Algorithmic Decision-Making — redrawn.



Unfairness, Bias, Discrimination

How the three words nest?



Unfairness

Quality of treating people unequally or in a way that is wrong or unreasonable.

* INEQUALITY

* PARTIALITY

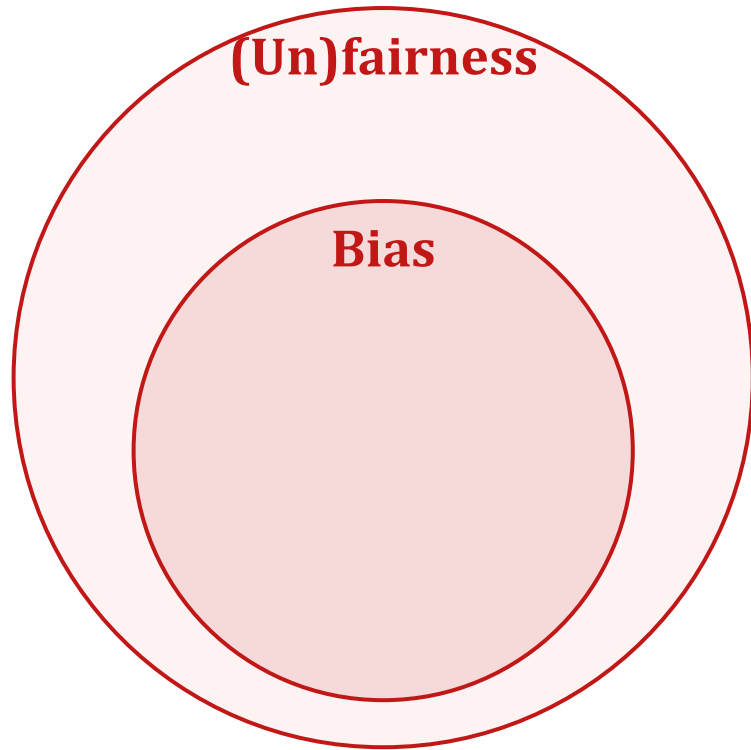
* INJUSTICE

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* INEQUALITY * PARTIALITY * INJUSTICE

Bias

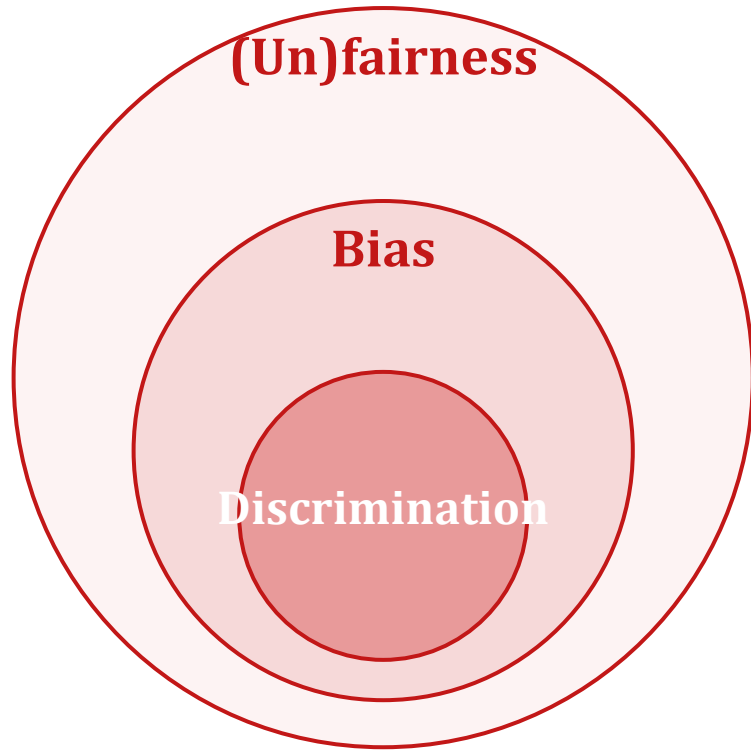
An unfair skew — on any basis: a protected trait, an arbitrary factor, a proxy.

After CDEI (2020), Review into Bias in Algorithmic Decision-Making — redrawn.



Unfairness, Bias, Discrimination

How the three words nest?



Unfairness

No skew required. An opaque process, or no way to appeal a decision, is unfair on its own.

Bias

An unfair skew — on any basis: a protected trait, an arbitrary factor, a proxy.

Discrimination

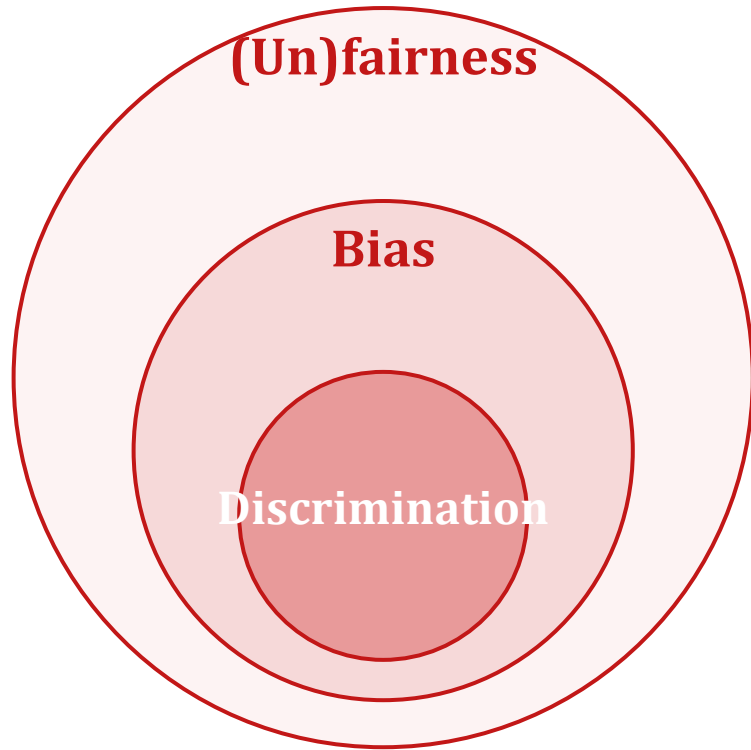
The legally actionable core: unfair treatment tied to a protected characteristic.

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An unfair skew — on any basis: a protected trait, an arbitrary factor, a proxy.

Discrimination

The legally actionable core: unfair treatment tied to a protected characteristic.

Removing bias does not make a system fair.

After CDEI (2020), Review into Bias in Algorithmic Decision-Making — redrawn.



Older Than Any Algorithm

“Fairness as a concept can be traced to the earliest schools of philosophy.”

Quotes: CDEI / Faculty (2020). Aristotle, *Nicomachean Ethics V*; Rawls, *A Theory of Justice* (1971).



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Fairness: Older Than Any Algorithm

“Fairness as a concept can be traced to the earliest schools of philosophy.”

Aristotle

c. 350 BCE

Equals treated equally;
unequals unequally — in
proportion to relevant
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“Justice as fairness”: rules you
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Politics & policy

today

“Along with justice, perhaps
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Historically an abstract quality, not a measurable one.

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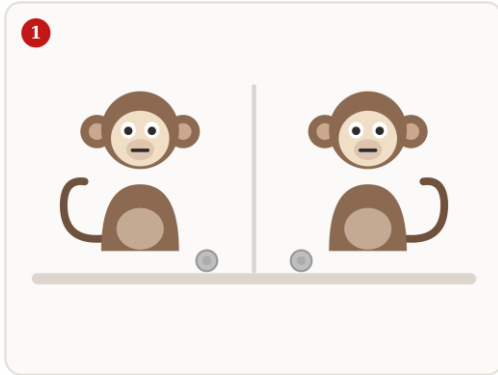
Is Fairness a Law of Nature?

**Is fairness a law of nature —
or something we made up?**



Is Fairness a Law of Nature?

Two capuchin monkeys, one experimenter, two kinds of reward.



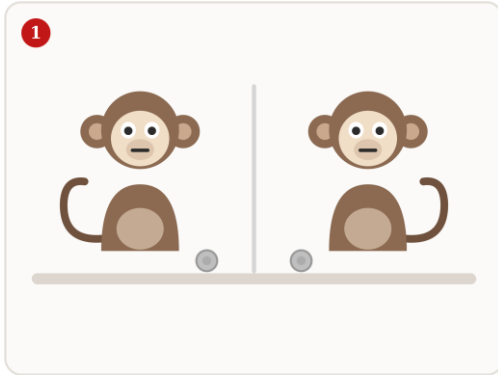
Both do the same task.

Brosnan & de Waal, "Monkeys reject unequal pay", Nature 425 (2003).

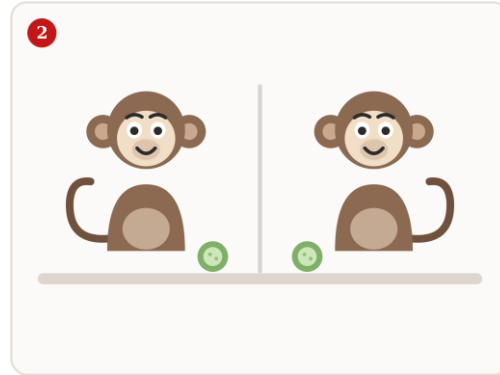


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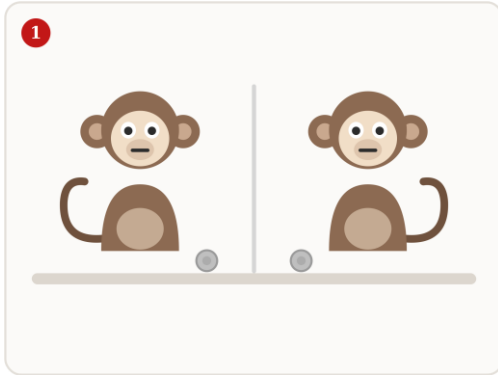
Both are paid a cucumber slice. Both content.

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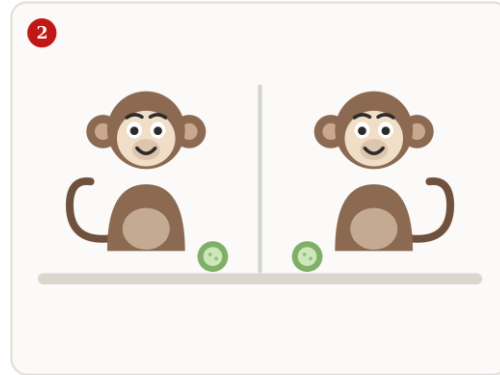


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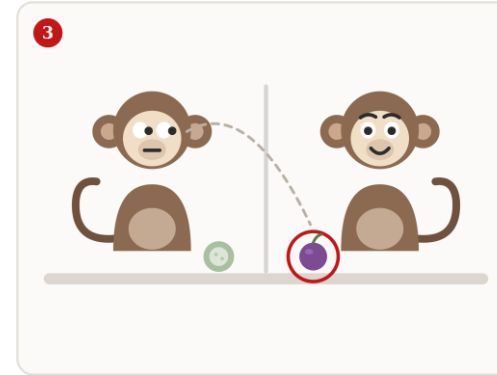
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Both do the same task.



Both are paid a cucumber slice. Both content.



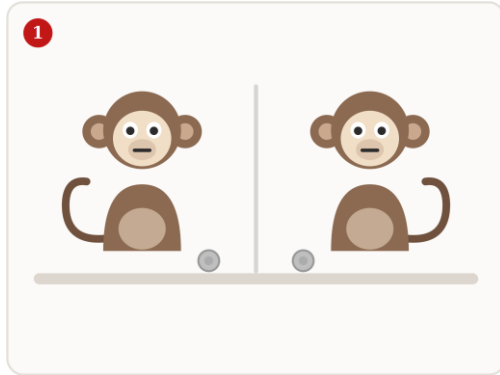
Same task — one is now paid a grape.

Brosnan & de Waal, "Monkeys reject unequal pay", Nature 425 (2003).

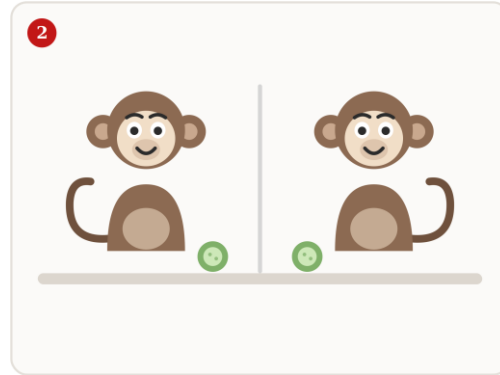


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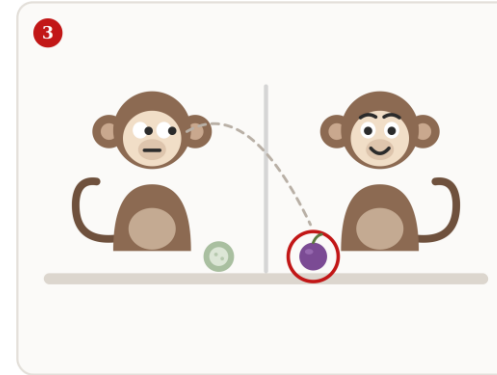
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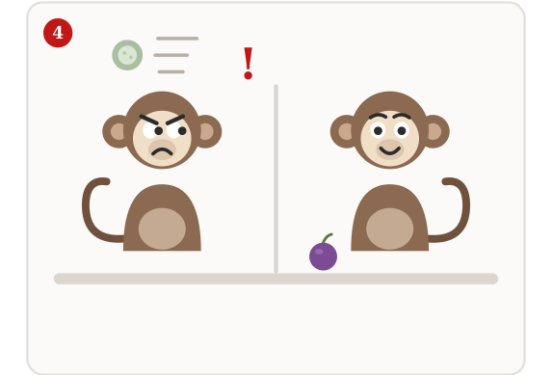
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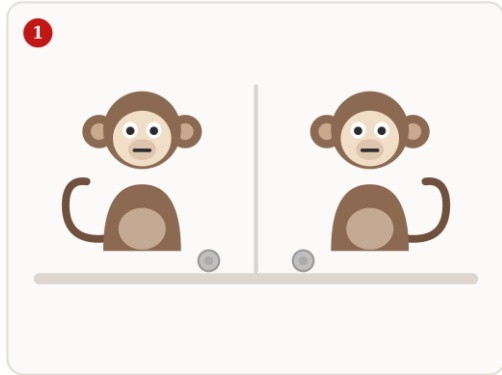
The other refuses the cucumber.

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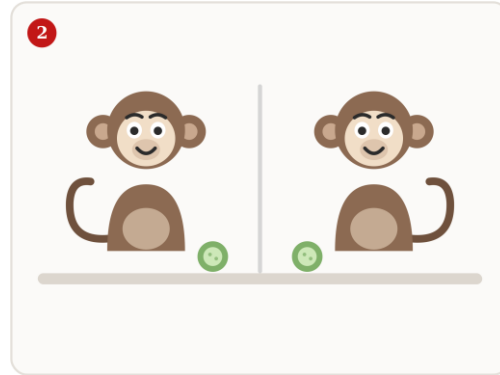


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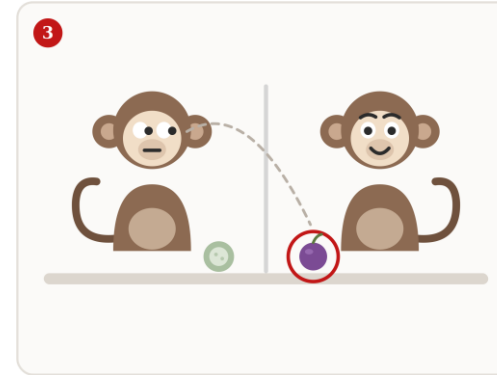
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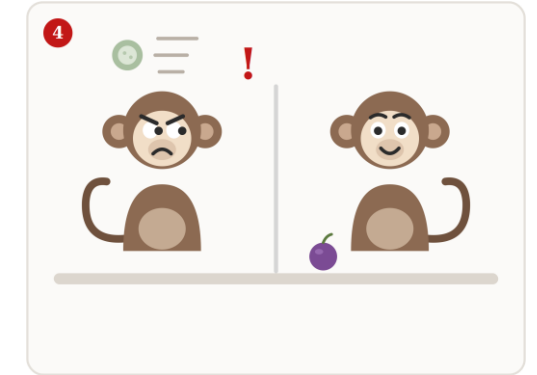
Both do the same task.



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Same task — one is now paid a grape.



The other refuses the cucumber.

Nature has the impulse.

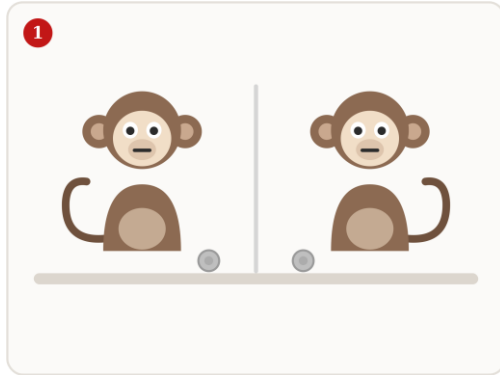
A sense of unequal pay — or frustration at a visible grape? Contested since 2003.

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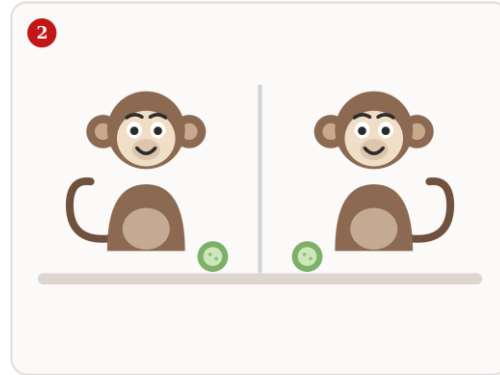


Is Fairness a Law of Nature?

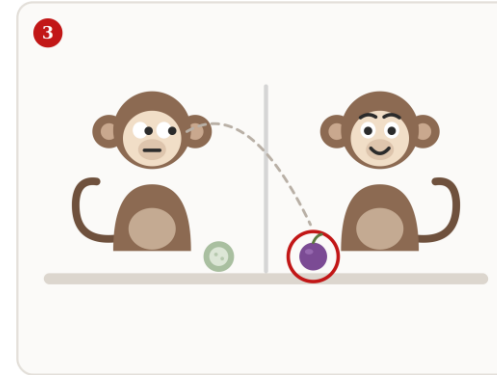
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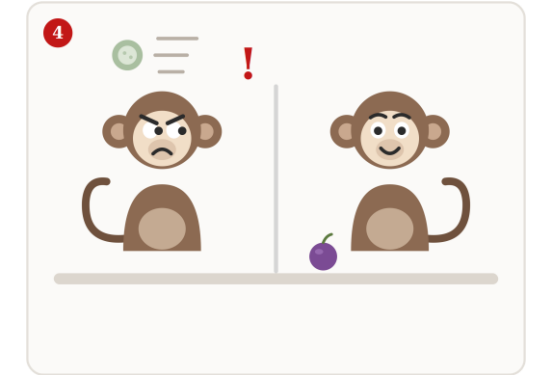
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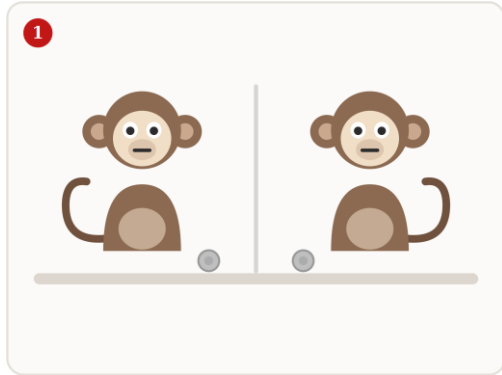
Trustworthy Large Language Models



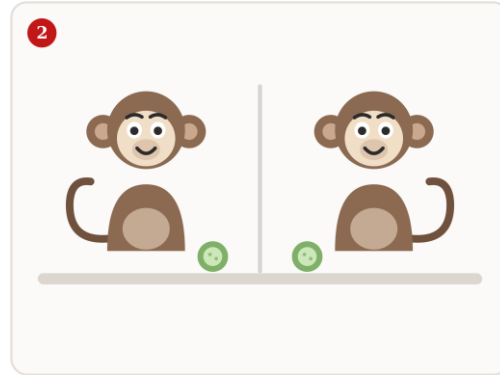
Yatin Nandwani

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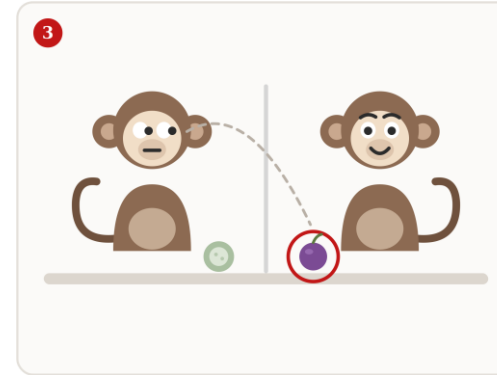
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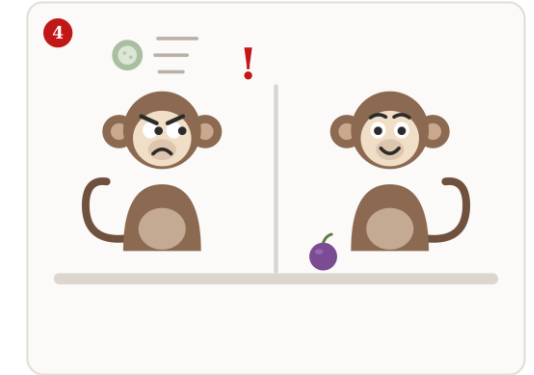
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Fairness is a man-made construct — and our notion of it keeps evolving.

Brosnan & de Waal, “Monkeys reject unequal pay”, Nature 425 (2003).



The Standard of Fairness Moves

Every one of these was, at the time, simply how a bank or a ballot worked.

1928

United Kingdom

Women vote on the same terms as men. Ten years earlier the age bar was 30 — and property-qualified.

Representation of the People Act 1928; Equal Credit Opportunity Act 1974; Equality Act 2010.



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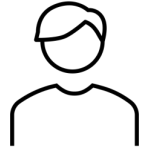
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What is considered “fair” keeps evolving.



Systems to Ensure Fairness



Societies have never trusted a decision-maker to be fair. They build procedure instead.

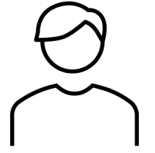
What we built

- A written rule, fixed before the case
- A stated reason for the decision
- A right to be heard, and to appeal
- Someone independent who can audit it

Procedural safeguards as characterised in CDEI (2020).



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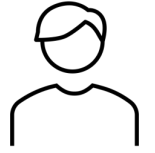
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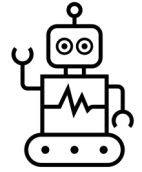
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Never foolproof — but someone decided, could be asked why, and could be overruled.

Procedural safeguards as characterised in CDEI (2020).



The Decision Moved to an Algorithm



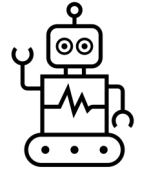
COMPAS

2016

Risk scores used in US pretrial decisions. Among defendants who did not go on to reoffend, ~45% of Black and ~23% of white defendants had been labelled high-risk.



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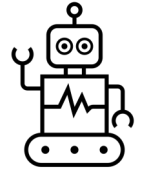
Amazon's CV screener

2014–17

Trained on ten years of the company's own hires. It learned to penalise the word “women’s” — as in women’s chess club captain. Scrapped.



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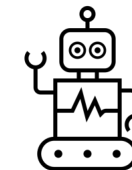
Workday

in court now

One screening service, many employers. The lawsuit in the headline you saw ten minutes ago.



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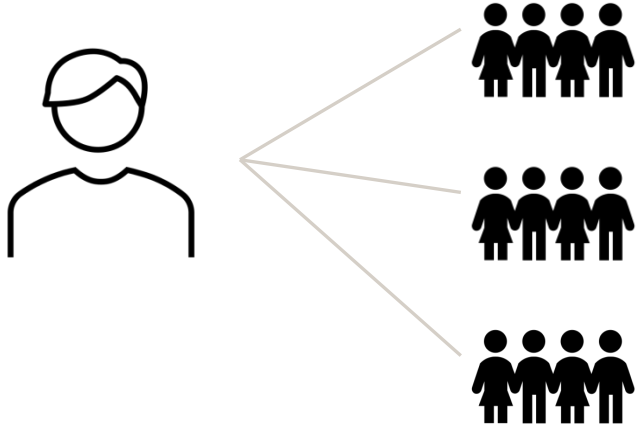
ProPublica (2016), figures via Chouldechova (2017); Reuters (2018), Amazon; Reuters (2026), Workday.

Not one of these is a language model. The problem is older than the technology.



One Person, or One Algorithm

What changes when the same skew moves from a desk into a deployment?

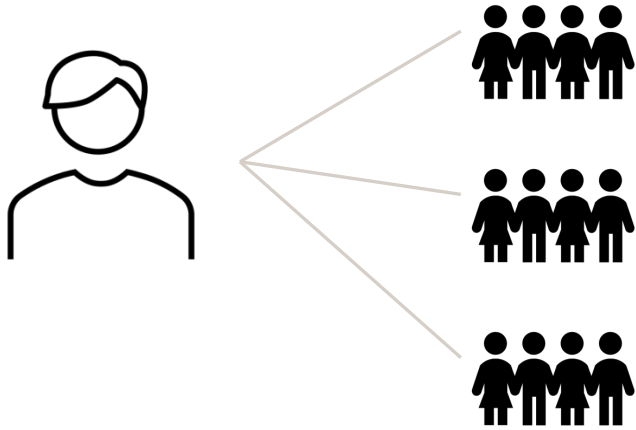


One desk. Hundreds of decisions.

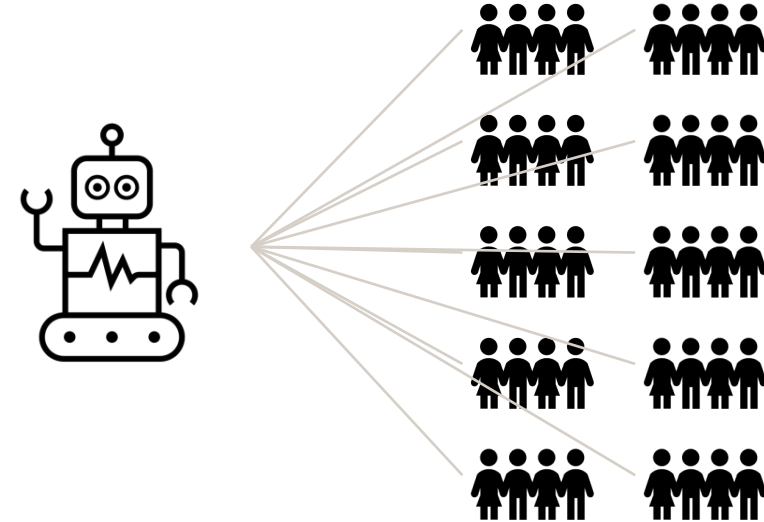


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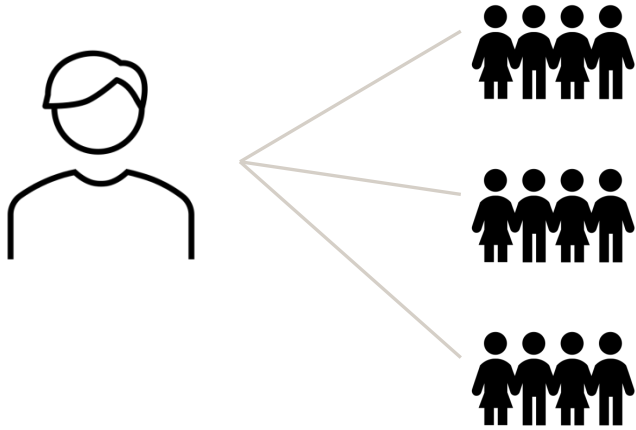


Every deployment. Millions of decisions.

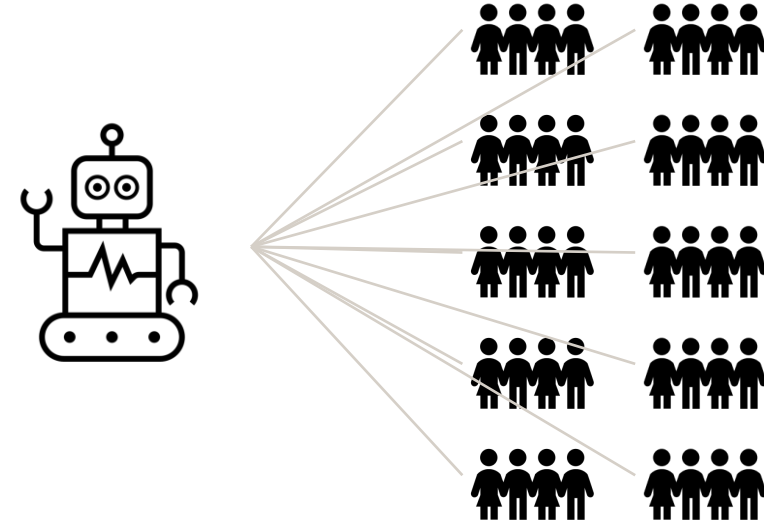


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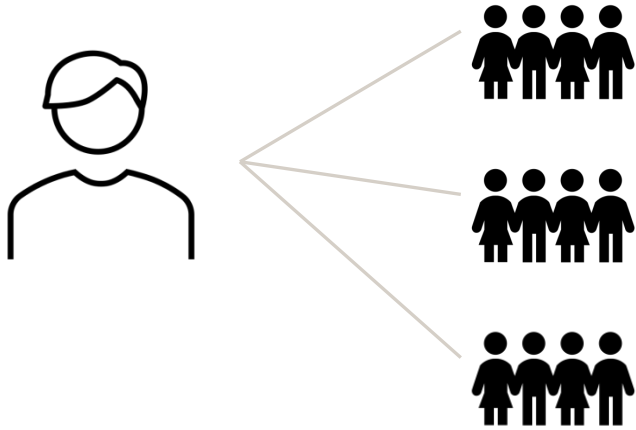
Reach

The same skew, applied everywhere the algorithm is used, at once.

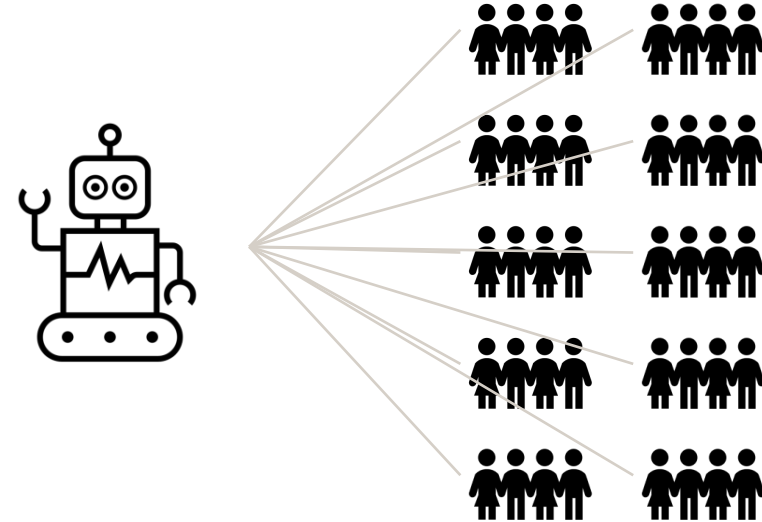


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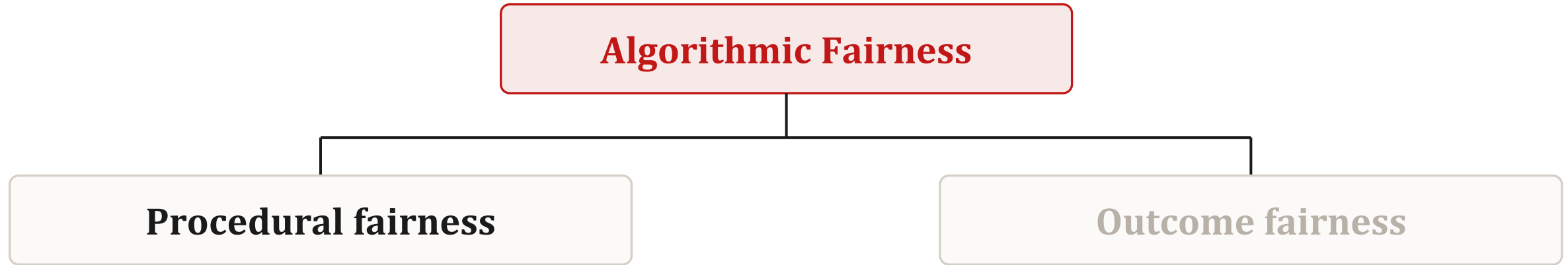
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**Extensive reach calls for a system to ensure algorithms are fair —
Algorithmic Fairness.**



Algorithmic Fairness: A Framework

The first question to ask of any decision-making algorithm.

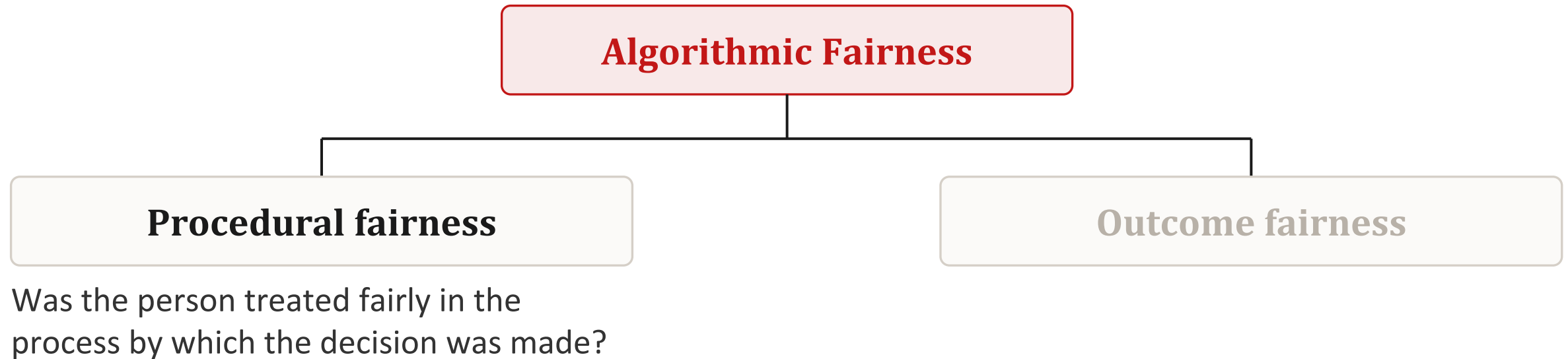


Organising frame after CDEI (2020), Review into Bias in Algorithmic Decision-Making.



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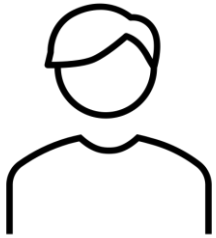


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Procedural Fairness

Fair treatment of the person within the process — not only inside the model.

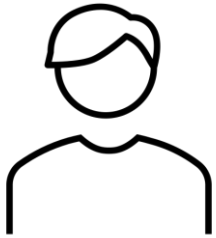


CDEI (2020): procedural fairness “encompasses wider elements of the decision-making process” than the algorithm’s design.



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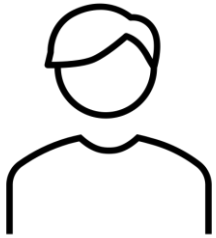
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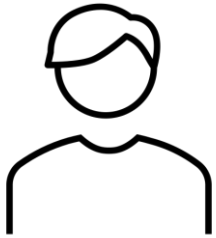
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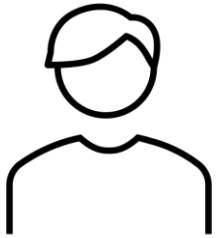
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The whole decision cycle in the real world — before, during and after the algorithm runs.

CDEI (2020): procedural fairness “encompasses wider elements of the decision-making process” than the algorithm’s design.



Fairness Through Unawareness

The first thing every engineer proposes: do not give the model the sensitive attributes.

income

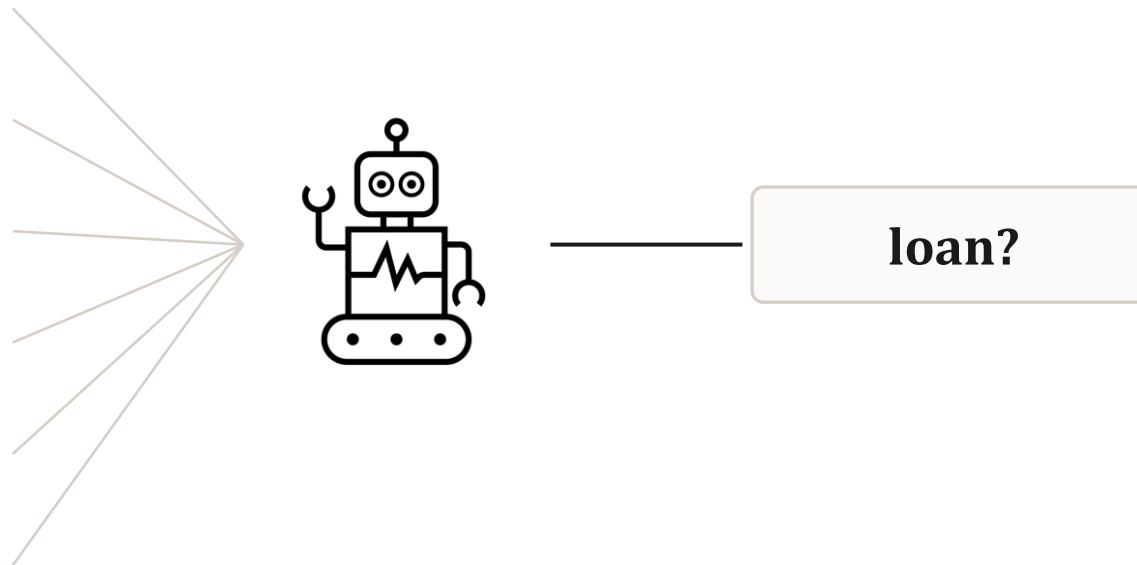
employment

existing debt

postcode

gender

race



CDEI (2020). Postcode–race is the review’s own example; Amazon’s screener never saw gender and learned “women’s”.



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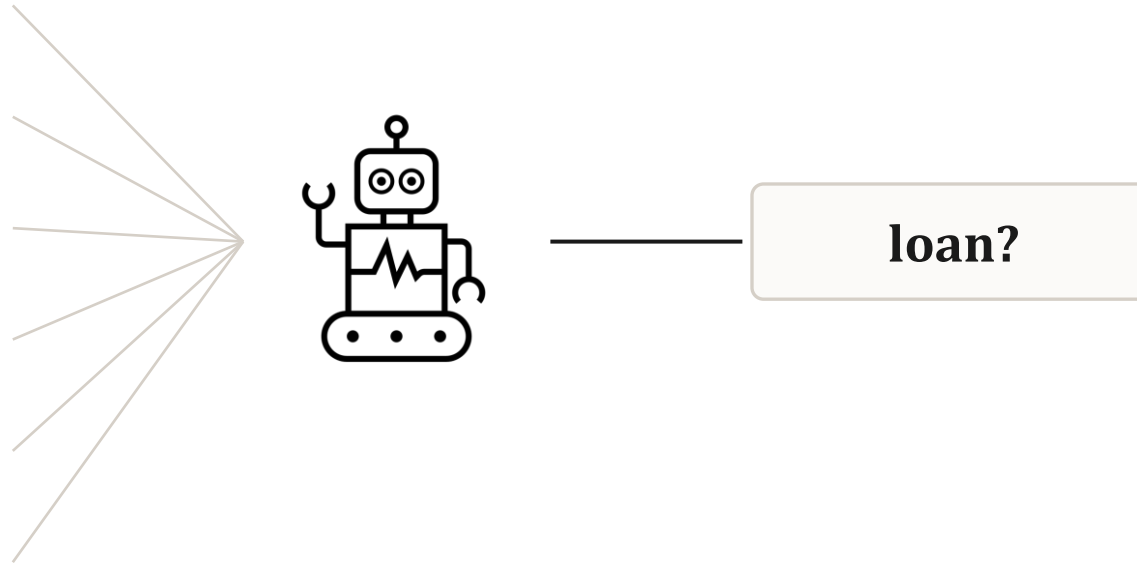
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The premise

If the model never sees gender or race, it cannot discriminate on them.

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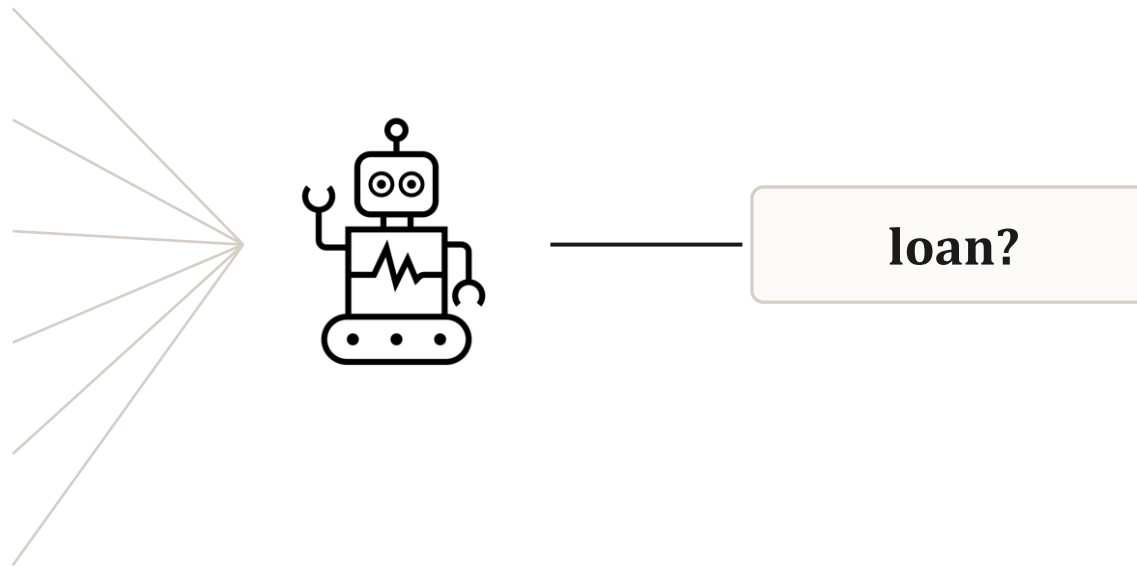
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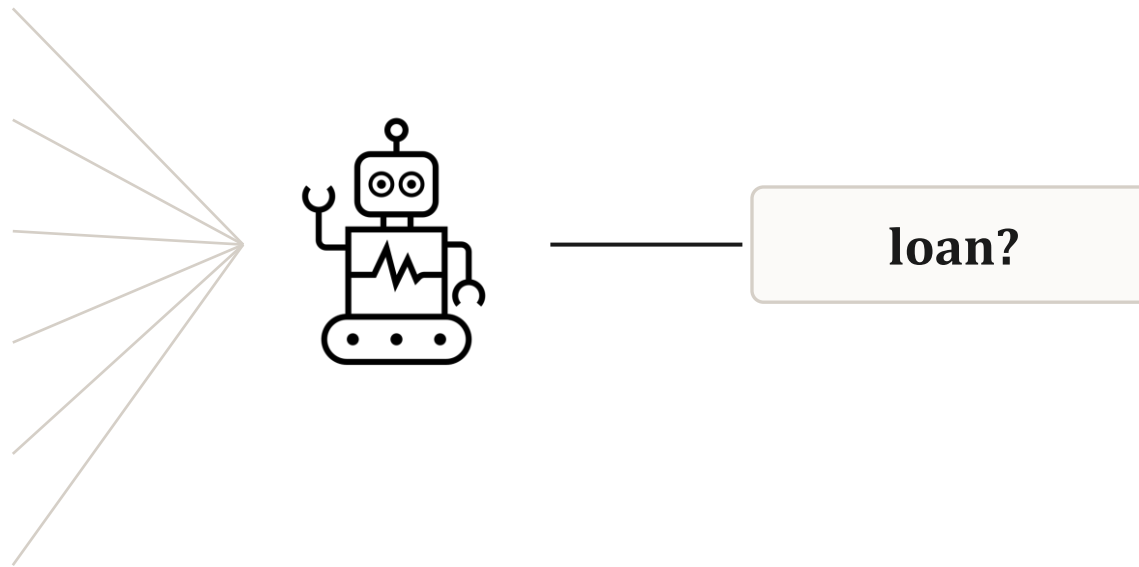
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Postcode can predict race. The model uses postcode. Nothing was checked at the output.

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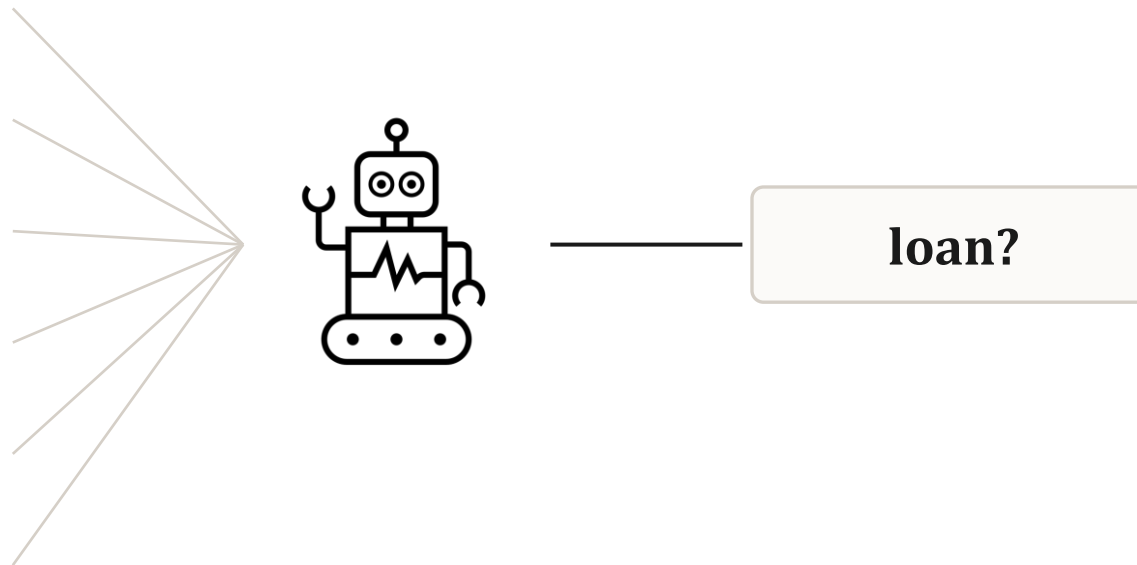
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Procedural: it constrains the inputs and says nothing about the outcome.

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Algorithmic Fairness: A Framework

The second question: not the process, the result.



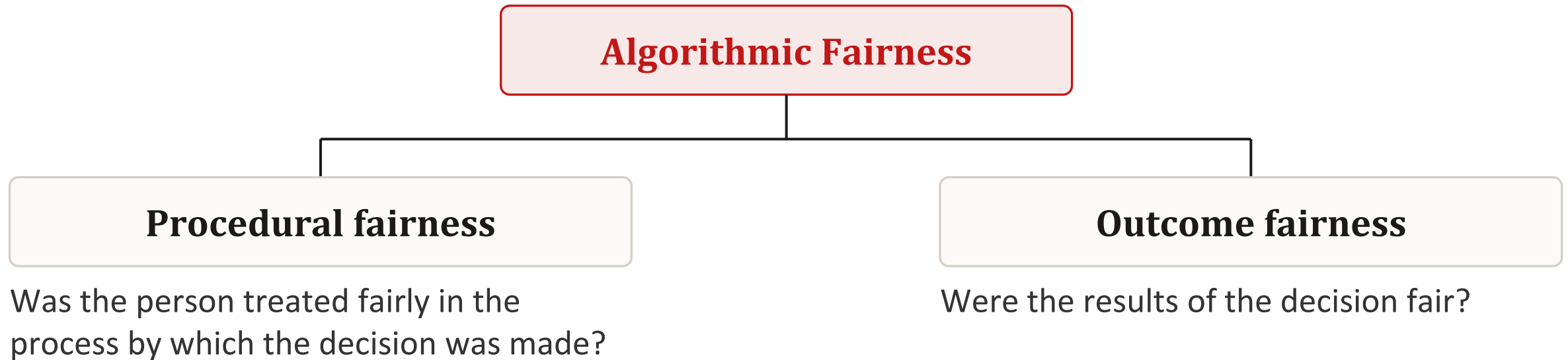
Was the person treated fairly in the process by which the decision was made?

Organising frame after CDEI (2020), Review into Bias in Algorithmic Decision-Making.



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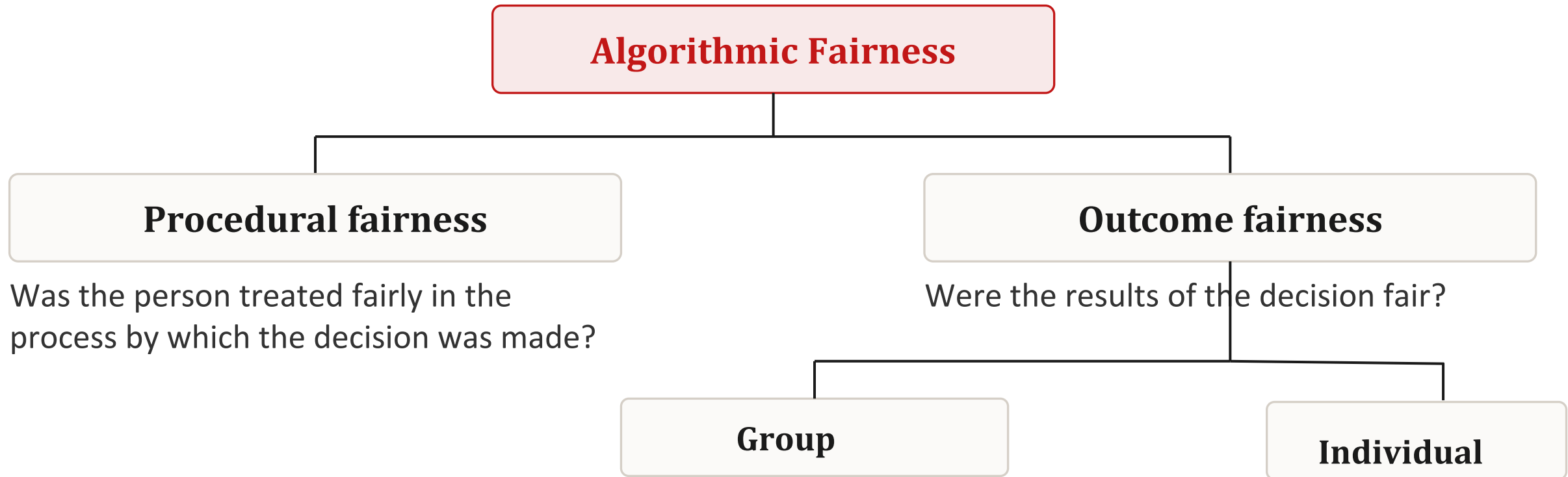


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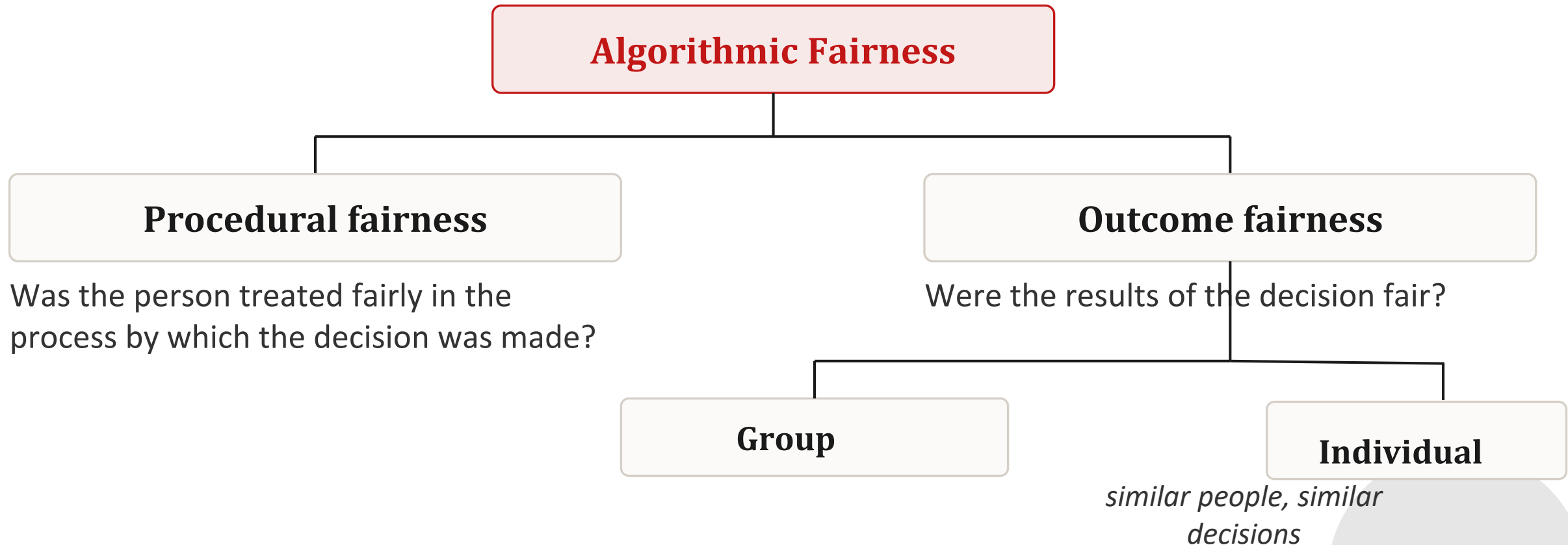


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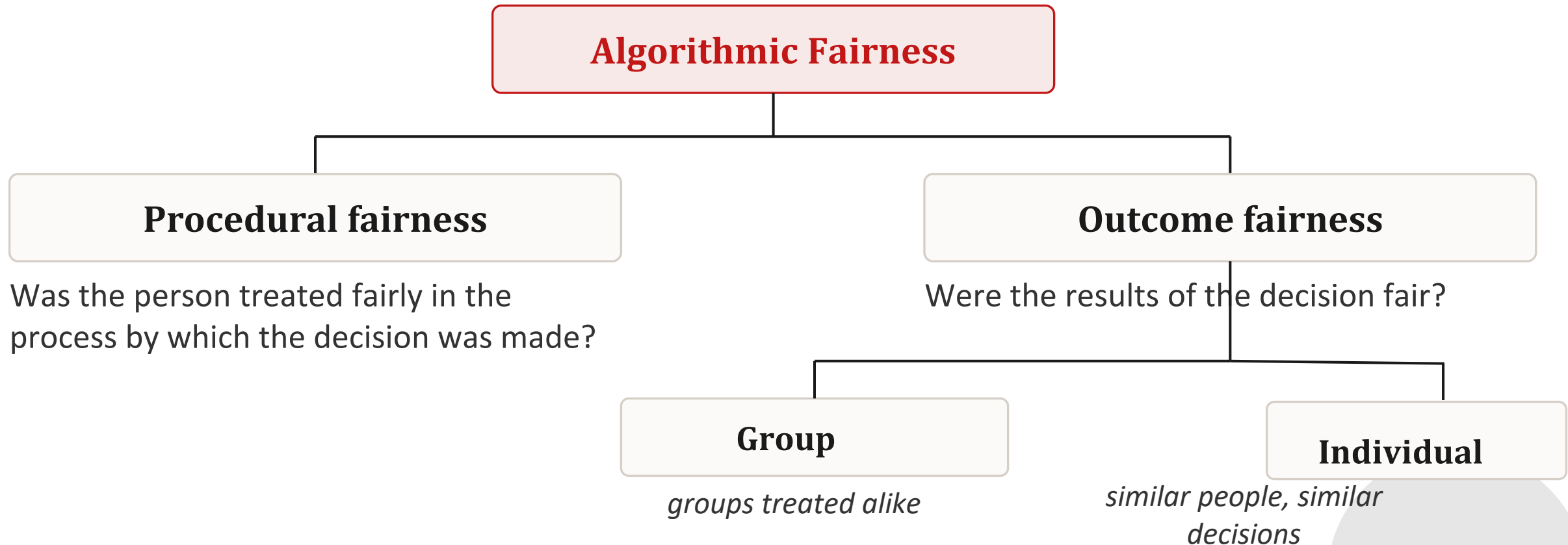


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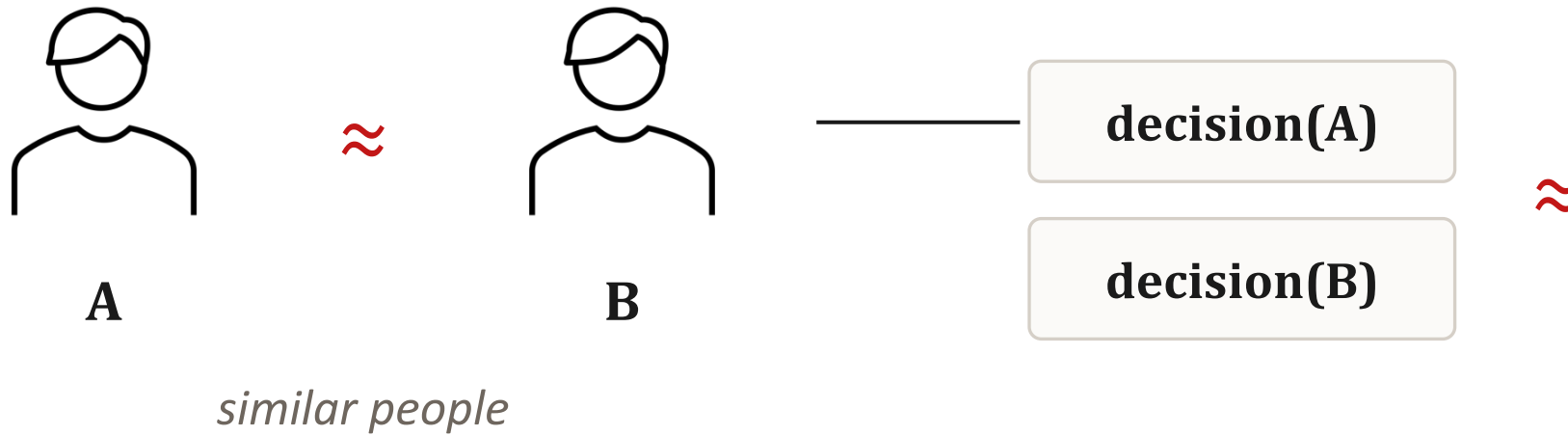


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Outcome Fairness: Individual

Similar individuals should receive similar decisions.

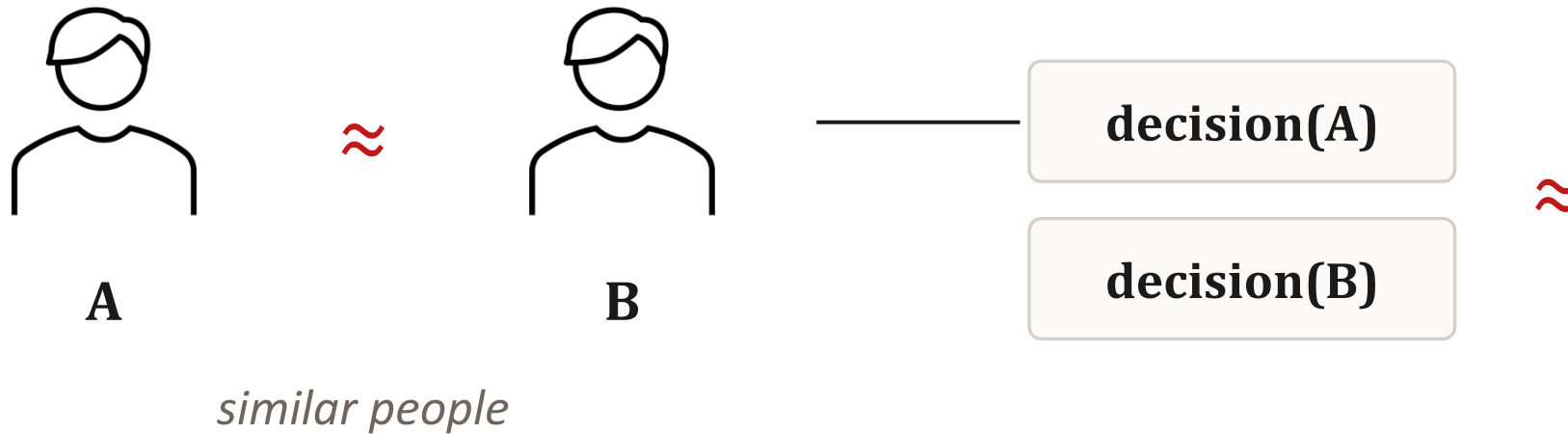


Dwork, Hardt, Pitassi, Reingold & Zemel, "Fairness Through Awareness" (ITCS 2012).



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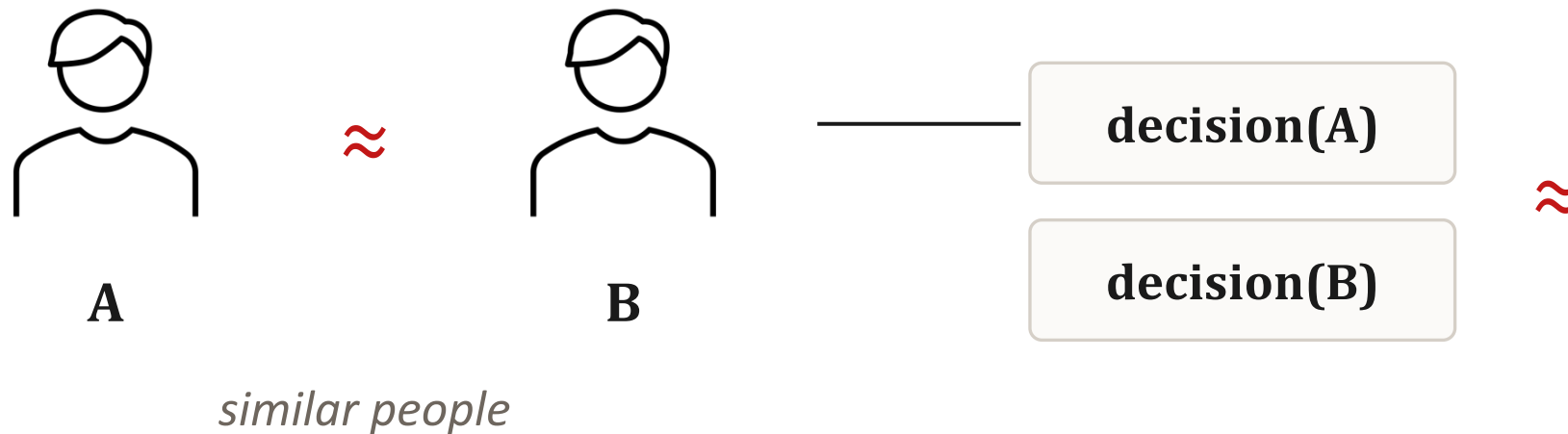
Similar — by what measure?

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Outcome Fairness: Individual

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Similar — by what measure?

Choosing the similarity measure is itself a judgement about which differences should matter. It reintroduces the problem it was meant to avoid.

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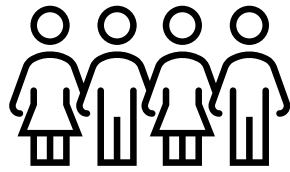


Outcome Fairness: Group

Different groups should be treated alike. Which groups?



G_1



G_2

Equality Act 2010, s.4. India, Art. 15: religion, race, caste, sex, place of birth.

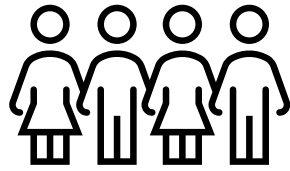


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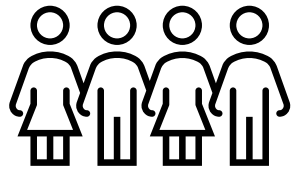


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Protected characteristics — UK Equality Act 2010

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gender reassignment

pregnancy and maternity

religion or belief

sexual orientation

disability

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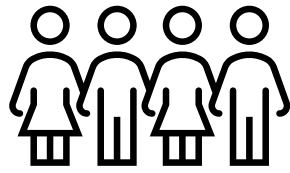


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For groups G_1 and G_2 , algorithm A should decide “similarly”. How do we operationalise that?

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Group Fairness: Demographic Parity

The first way to operationalise “decides similarly”.

$$P(\text{decision} = \text{yes} \mid G_1) = P(\text{decision} = \text{yes} \mid G_2)$$

the same acceptance rate — or the same error rate — in both groups

Also called independence / statistical parity. Buolamwini & Gebru (2018) on face analysis.



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Where it does not

Loan approval. Gender should play no role — but if repayment rates differ between groups, even a perfect predictor fails this test.

Also called independence / statistical parity. Buolamwini & Gebru (2018) on face analysis.



Group Fairness: Demographic Parity

The first way to operationalise “decides similarly”.

$$P(\text{decision} = \text{yes} \mid G_1) = P(\text{decision} = \text{yes} \mid G_2)$$

the same acceptance rate — or the same error rate — in both groups

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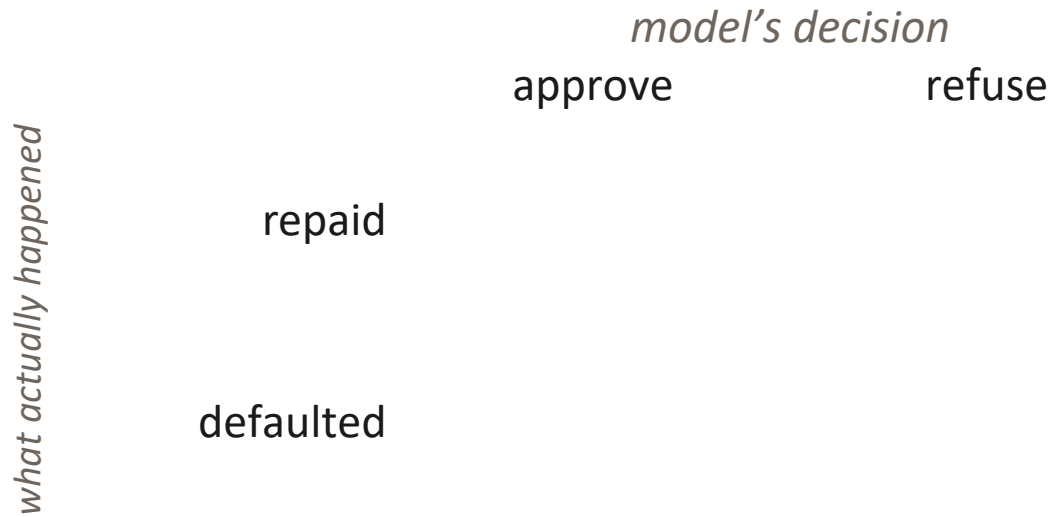
So the definition must take into account what actually happened.

Also called independence / statistical parity. Buolamwini & Gebru (2018) on face analysis.



Group Fairness: Equalised Odds

Loan approval, with hindsight: who actually repaid, and who was approved.



Hardt, Price & Srebro, NeurIPS 2016. Also called separation.



Group Fairness: Equalised Odds

Loan approval, with hindsight: who actually repaid, and who was approved.

		<i>model's decision</i>	
		approve	refuse
<i>what actually happened</i>	repaid	TP	FN
	defaulted	FP	TN

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True-positive rate

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

of those who would repay, the fraction approved

False-positive rate

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

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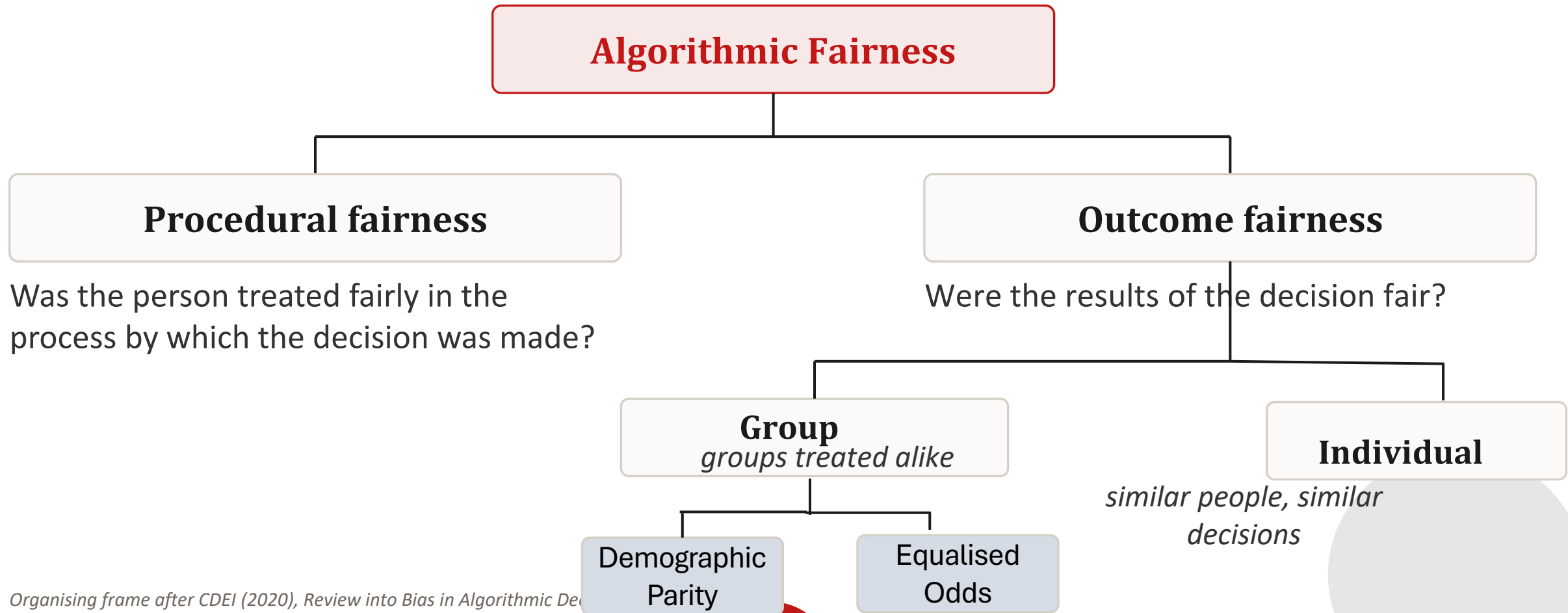
$$\text{TPR}(G_1) = \text{TPR}(G_2) \quad \text{and} \quad \text{FPR}(G_1) = \text{FPR}(G_2)$$

Equal chances of a right — and a wrong — decision in every group.

Hardt, Price & Srebro, NeurIPS 2016. Also called separation.



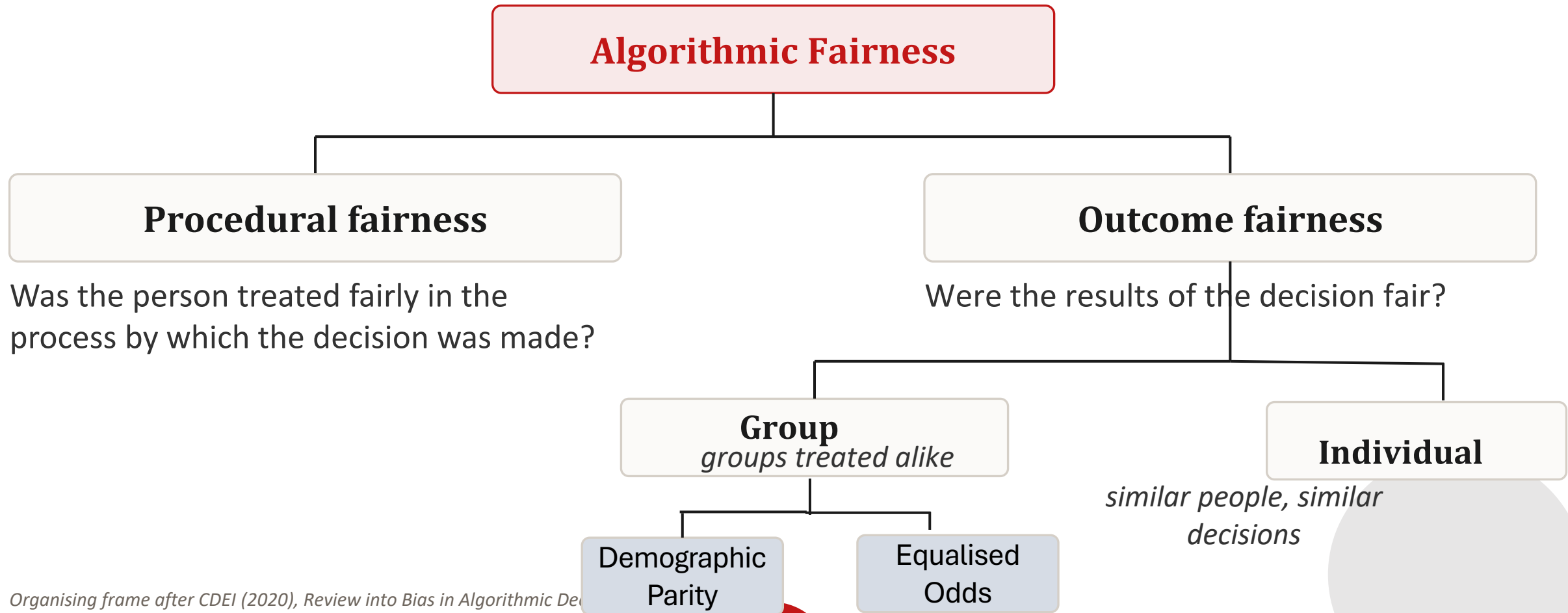
Algorithmic Fairness: A Framework



Organising frame after CDEI (2020), Review into Bias in Algorithmic Decision Making



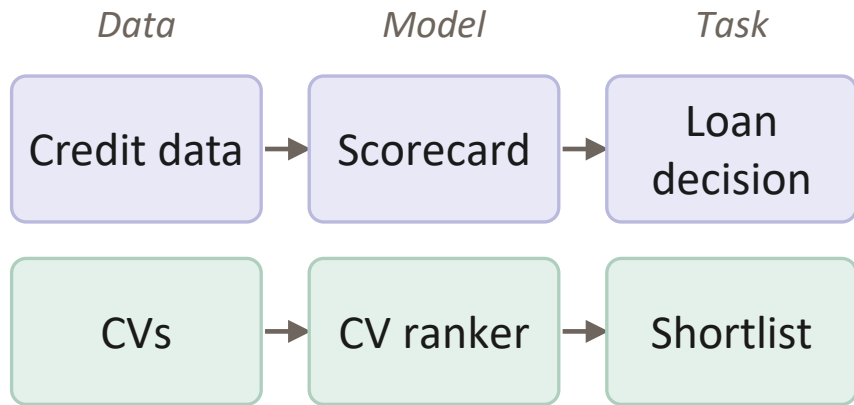
Algorithmic Fairness: In the Era of LLMs



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Algorithmic Fairness: In the Era of LLMs

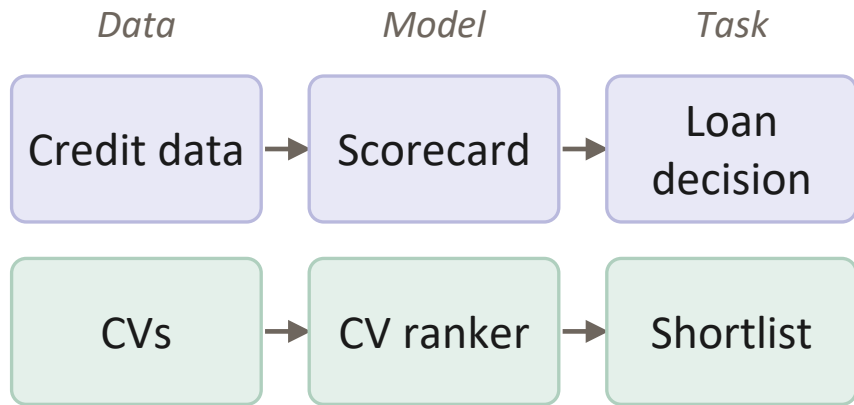


Before

One model, one decision, one audit.

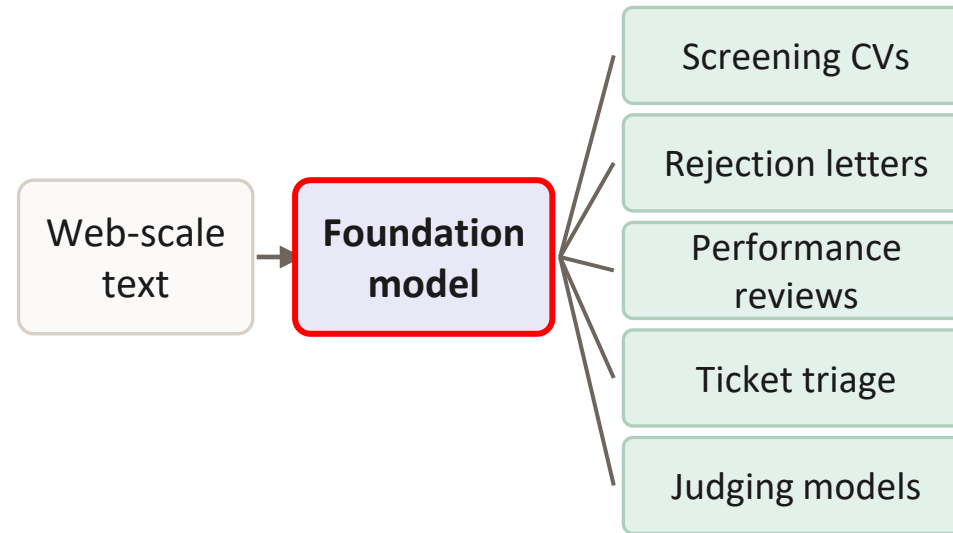


Algorithmic Fairness: In the Era of LLMs



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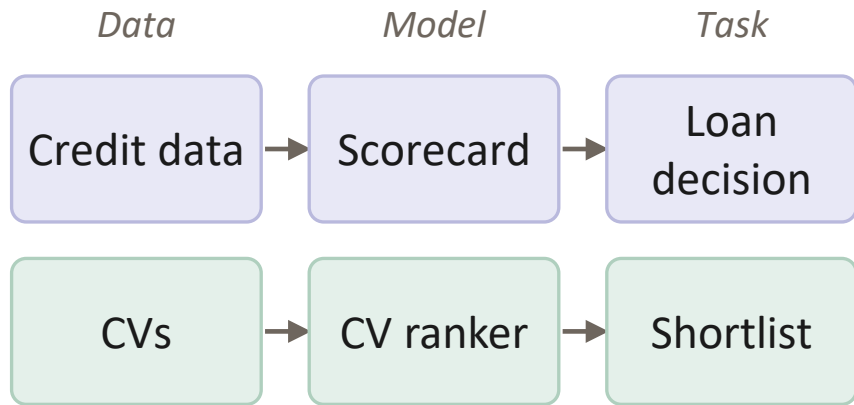


Now

One model, every decision. One skew, everywhere.



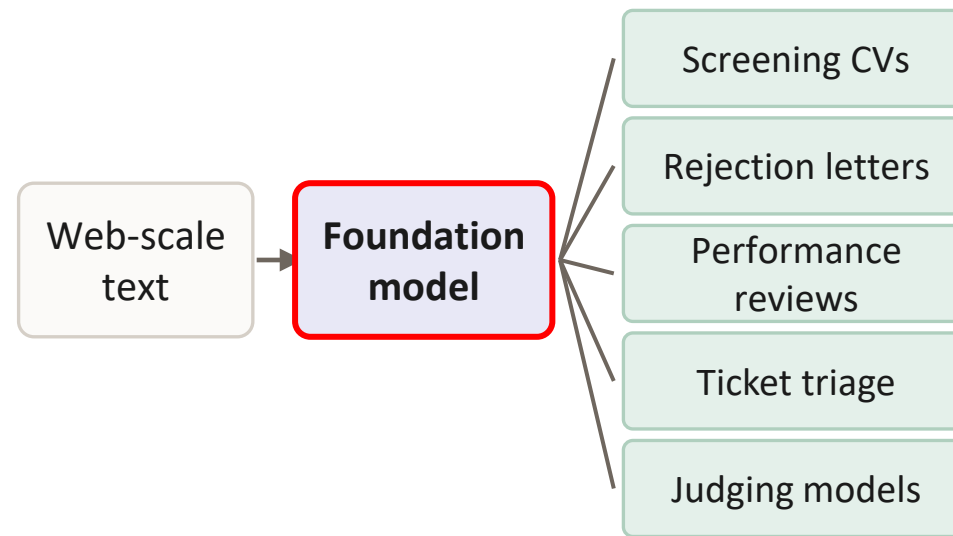
Algorithmic Fairness: In the Era of LLMs



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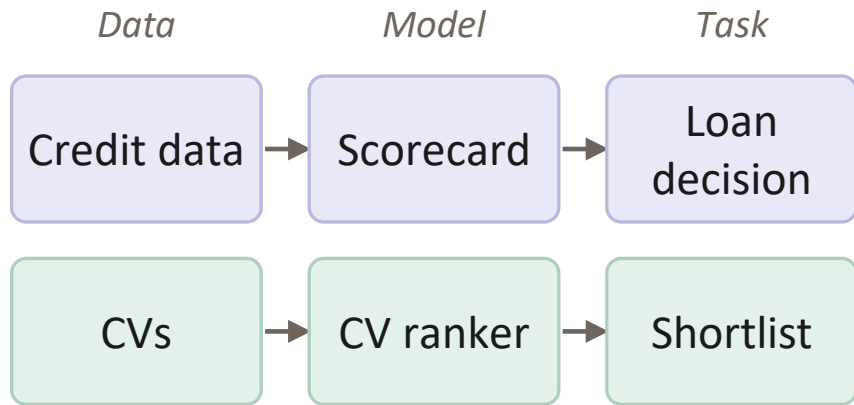


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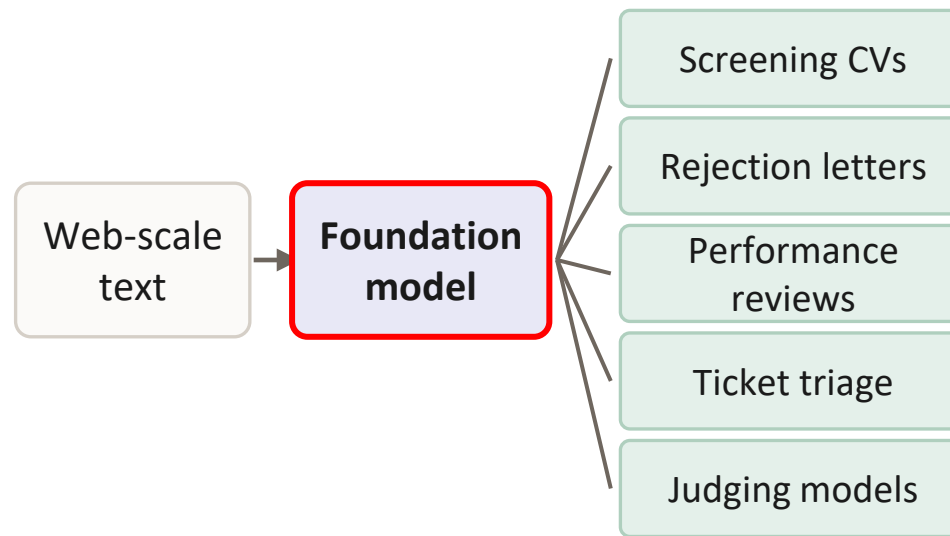
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- The model is now the judge — bias in the evaluator is invisible to every metric built on its scores

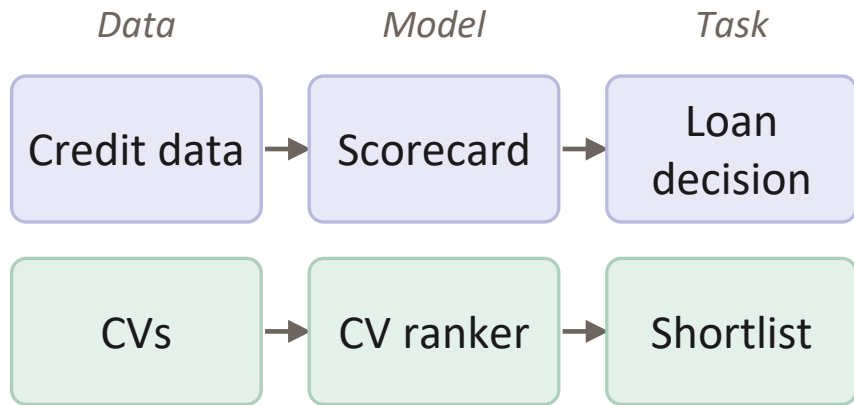


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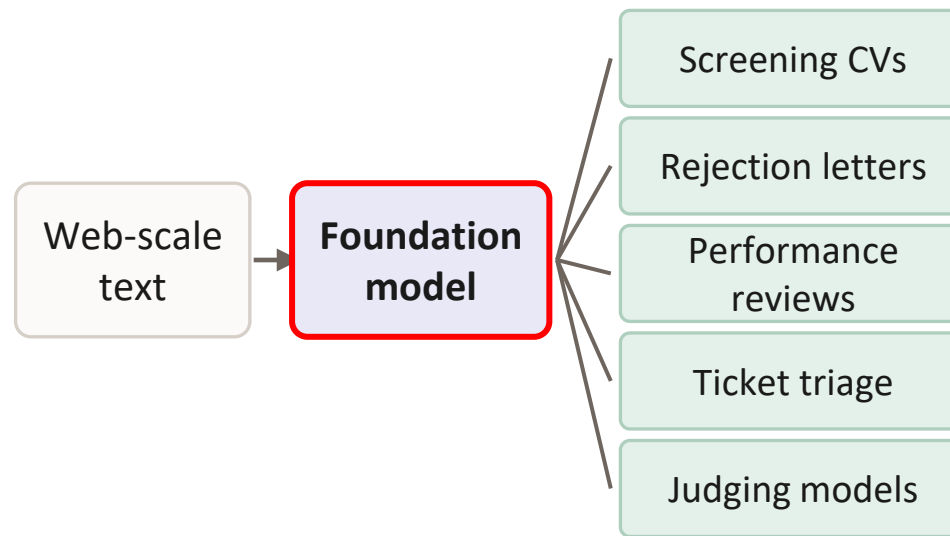


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One model, one decision, one audit.

- One model, every decision — a single skew propagates into every downstream use at once
- The model is now the judge — bias in the evaluator is invisible to every metric built on its scores
- Nobody sees the decision boundary — no coefficients to audit, just weights and a prompt

The decision layer is now a language model.



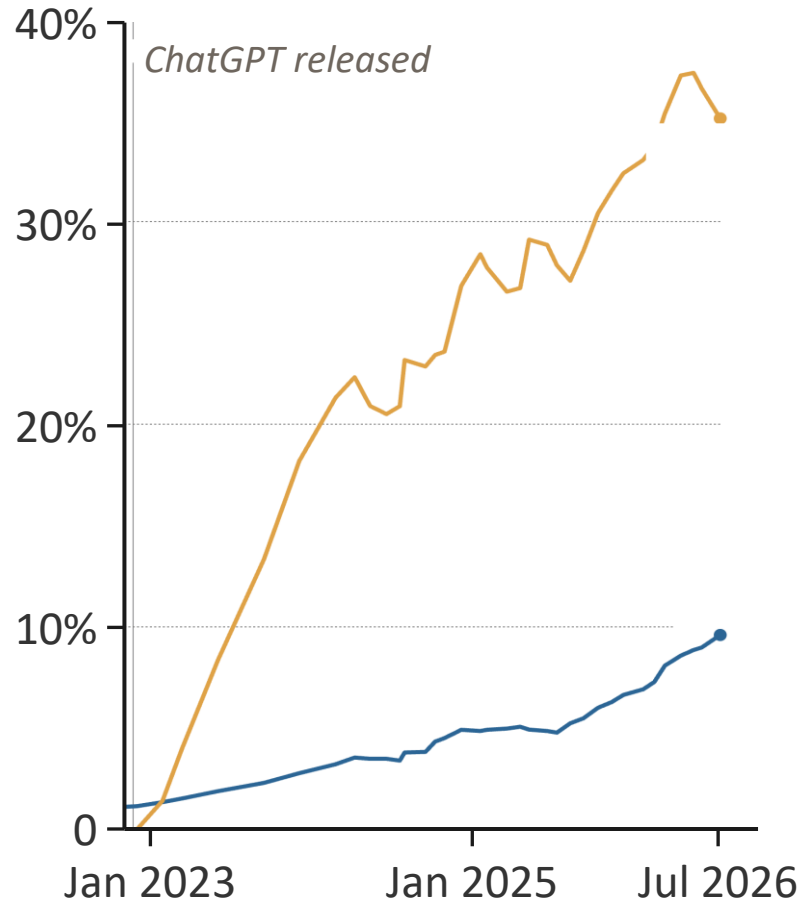
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Not Only Decisions

Share of English web pages showing signs of AI editing or authorship.

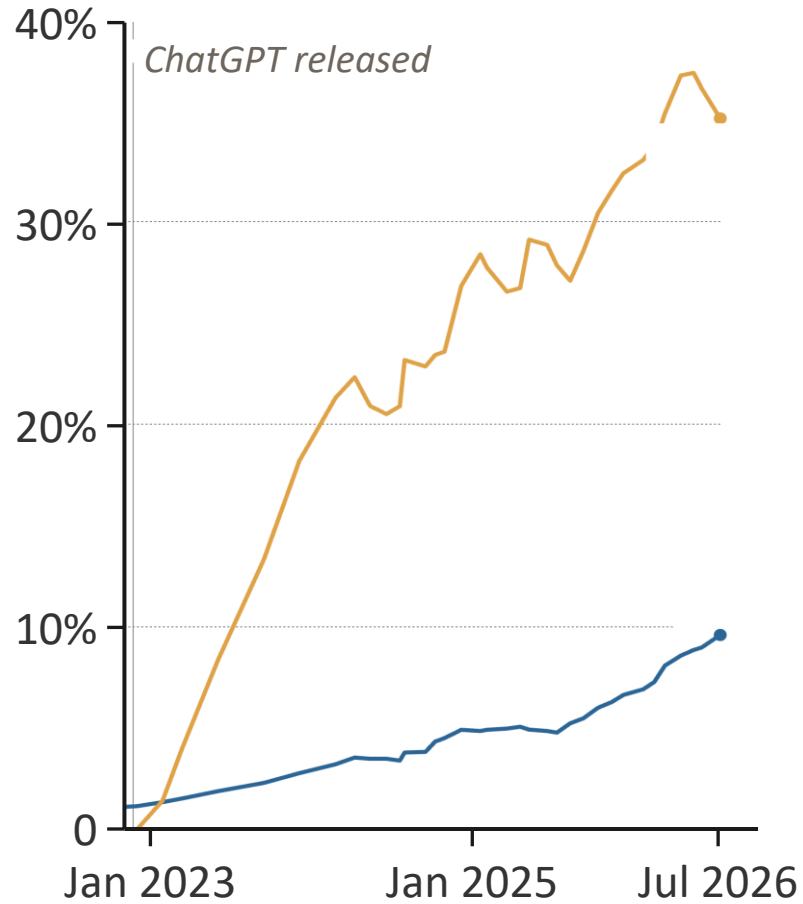


Pew Research Center (2026): 490,000 English pages from Common Crawl, Jan 2021 – Jul 2026, scored with Pangram.
Curves are the published chart; axes and labels redrawn.



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■ 35%

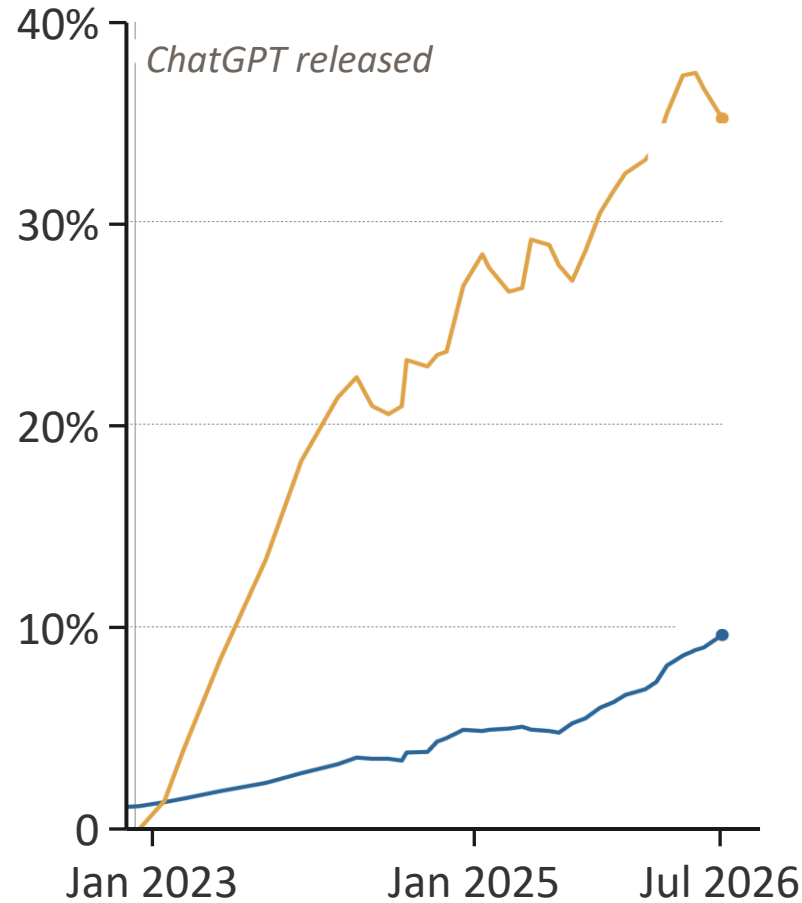
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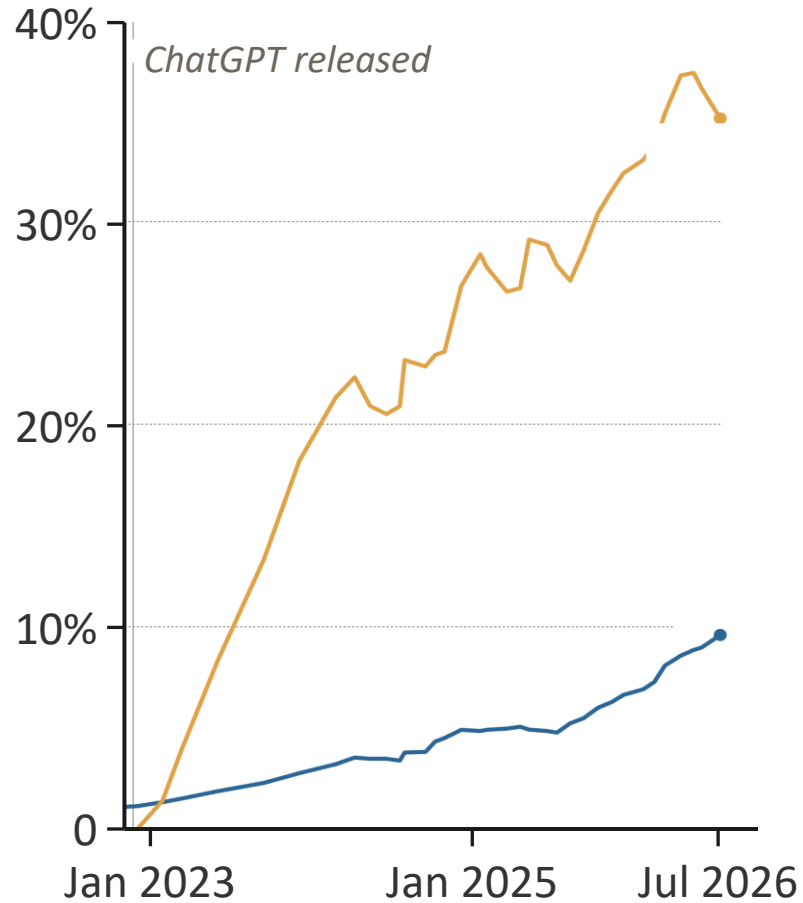
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It is text that people read.

The output is no longer only a decision.

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Two Families of Harm

What kind of damage does the skew do? The two halves need different tools.

REPRESENTATIONAL

How groups are portrayed.

ALLOCATIONAL

How resources and opportunities are distributed.

Gallegos et al. (2024), Table 1 — after Crawford, “The Trouble with Bias,” NeurIPS 2017 keynote



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Derogatory language — slurs, pejoratives

Disparate performance — works worse for some

Erasure — a group absent from the output

Exclusionary norms — one group as default

Misrepresentation — a distorted picture

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ALLOCATIONAL

How resources and opportunities are distributed.

Direct discrimination — worse treatment because of a protected characteristic

Indirect discrimination — a neutral rule that disadvantages a group, via proxies

Two categories against seven — and yet almost the entire fairness-metrics literature you just met was built for this column.

Gallegos et al. (2024), Table 1 — after Crawford, “The Trouble with Bias,” NeurIPS 2017 keynote



Why That Split Is the Hinge

What every criterion so far required — and what a sentence does not have.



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That apparatus is Lecture 2.



Sources of Bias in LLMs

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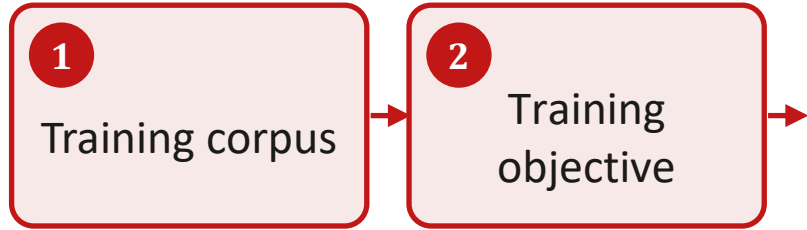
Training corpus →

1 • Training corpus

The web is not a sample of humanity: over- and under-representation, and filtering nobody documented.



Sources of Bias in LLMs



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2 • Training objective

The loss rewards fitting co-occurrence — and amplifies it. Nothing in it opposes bias.



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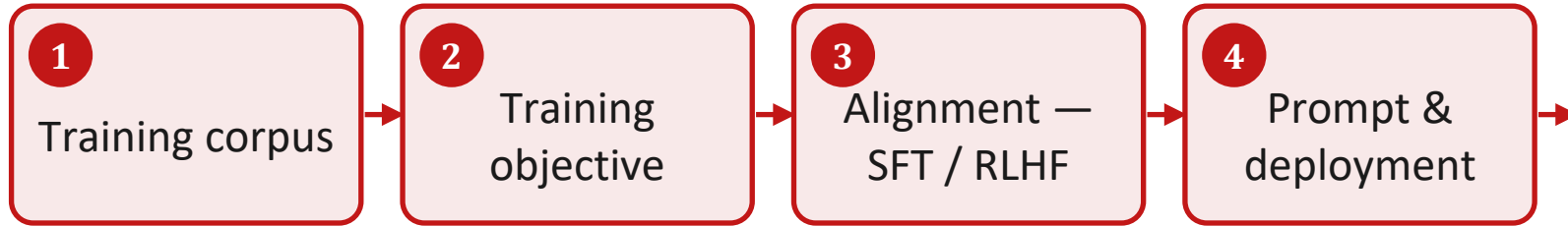
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3 • Alignment — SFT / RLHF

A narrow annotator pool. Optimising to an average preference can collapse minority views.



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System prompt, persona, in-context examples, option order. None of it is in the weights.



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5 • Evaluation & feedback

Benchmarks encode their authors' assumptions — and output re-enters tomorrow's corpus.

Five entry points — five different fixes.



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Size Doesn't Guarantee Diversity

Common Crawl is “petabytes of data collected over 8 years.” Surely that is everyone?

Four filters narrow the population — all of them before you write a line of training code.

1 Who has internet access Over-represents younger users and developed countries.



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A larger sample of a narrowed population is still a sample of a narrowed population.



Built for English, Used on Everyone

The population whose text trains the model is not the population it is used on.

Whose text trains it

- English, and a few other high-resource languages
- “Similar large-scale models aren’t being produced for Dhivehi or Sudanese Arabic”
- Written by whoever was online, posting, and allowed to stay

Bender, Gebru, McMillan-Major & Shmitchell (2021), “On the Dangers of Stochastic Parrots”, §2 and §4.1.



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Selection bias: training population $\neq$ deployment population.

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And Then We “Clean” It

The last narrowing is the one we perform deliberately, believing it to be hygiene.

GPT-3

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Cleaning is a value judgement, applied at scale and undocumented.

Bender et al. (2021), §4.



How Would We Show the Bias Is in the Text?

We cannot interview a corpus. But the corpus left something behind we can measure.

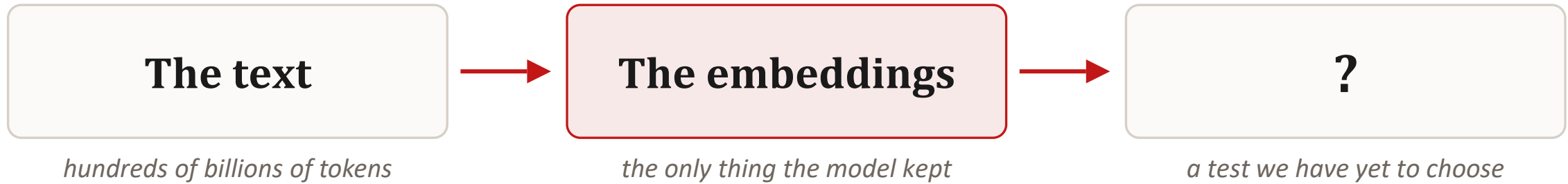
The text

hundreds of billions of tokens



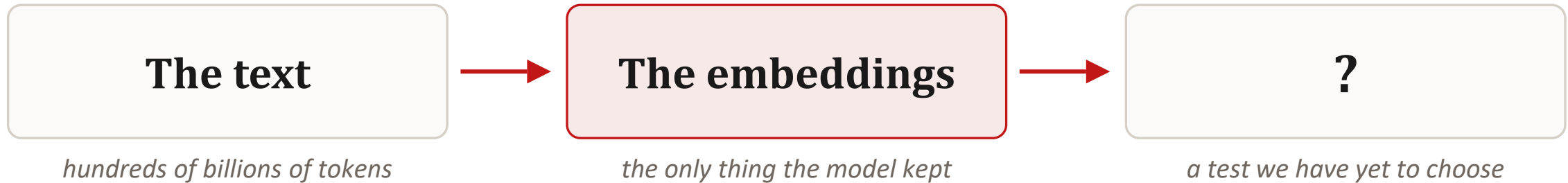
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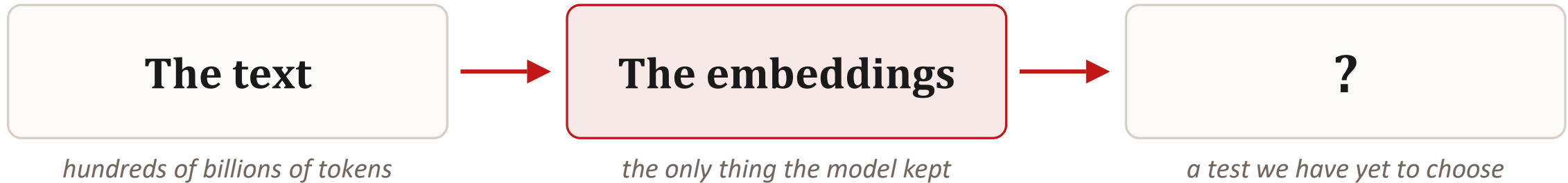


Whatever regularity the text carried – is it visible in the embeddings?



How Would We Show the Bias Is in the Text?

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Whatever regularity the text carried – is it visible in the embeddings?

So: probe the embeddings. With what instrument?



How Do We Test Implicit Bias in People?

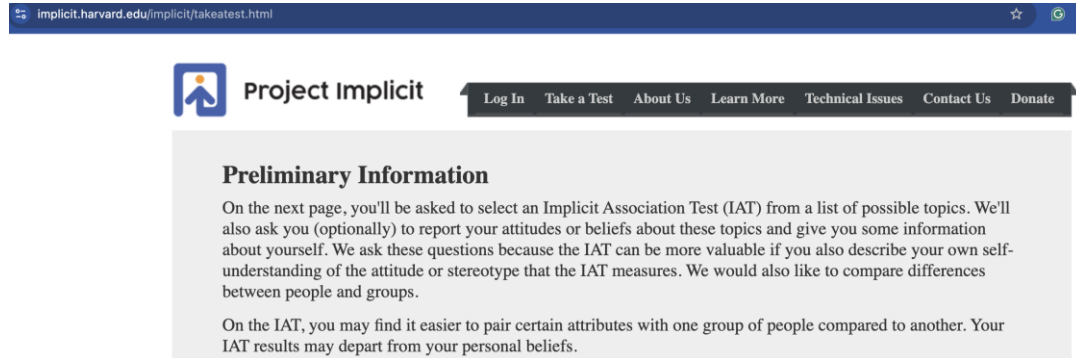
The same problem: you cannot get at it by asking.

Screenshots: implicit.harvard.edu. The test: Greenwald, McGhee & Schwartz (1998).



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Harvard's public site has run the Implicit Association Test online since 1998, across a menu of topics — race, gender and science, age, disability.

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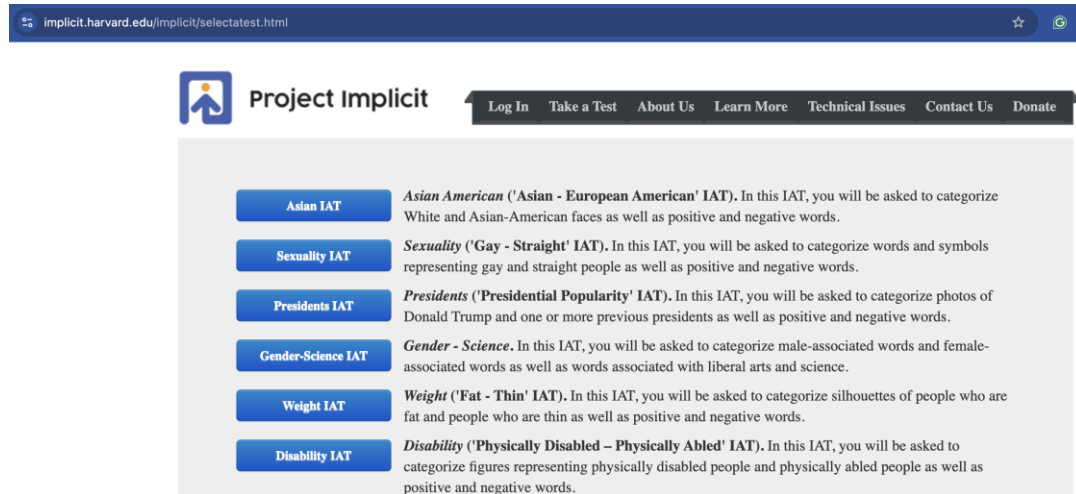
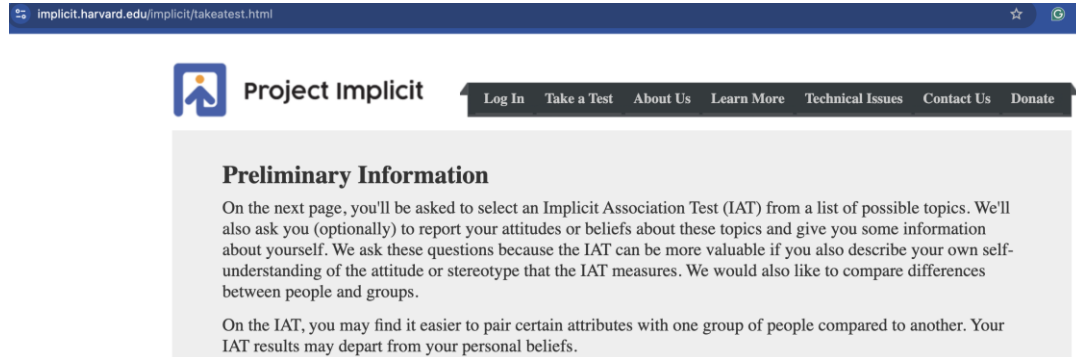
Trustworthy Large Language Models



Yatin Nandwani

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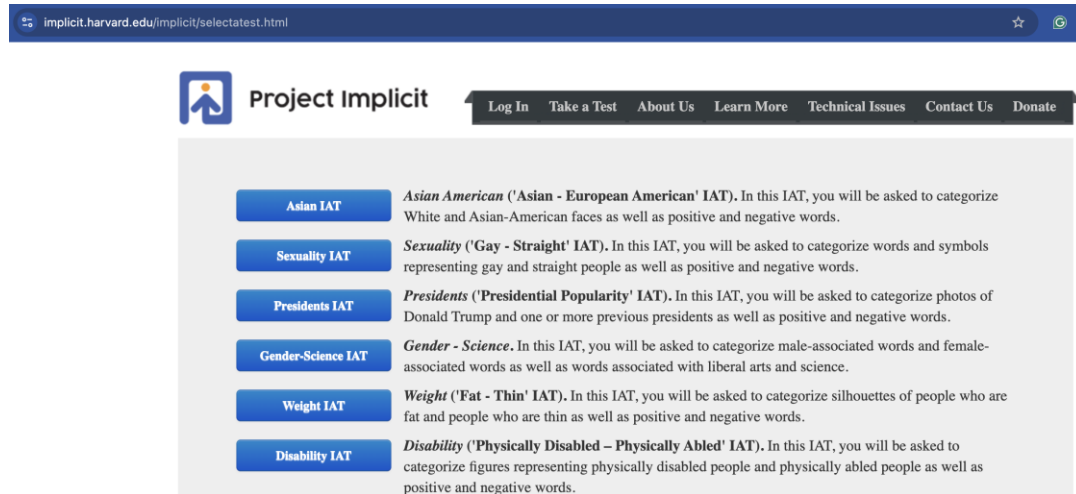
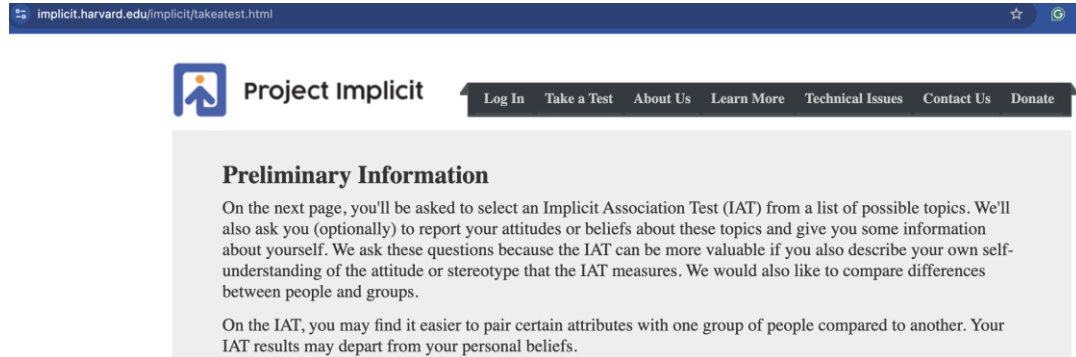
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An association a person will not report — or does not know they hold — cannot be reached by a questionnaire. So measure something they cannot control: how fast they respond.



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Reaction time is the measure.



The Implicit Association Test

Sort words into groups using two keys, as fast as you can.

Target groups

Male

Ben · Paul · Daniel · John · Jeffrey

Female

Rebecca · Michelle · Emily · Julia · Anna

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Career · Corporation · Salary · Office · Professional · Business

Family

Wedding · Marriage · Parents · Relatives · Home · Children

Two target groups, two attributes, two keys — and five blocks of trials.

Greenwald, McGhee & Schwartz (1998).



Trustworthy Large Language Models



Yatin Nandwani

The Implicit Association Test

Sort words into groups using two keys, as fast as you can.

Target groups

Male

Ben · Paul · Daniel · John · Jeffrey

Female

Rebecca · Michelle · Emily · Julia · Anna

Attributes

Career

Career · Corporation · Salary · Office · Professional · Business

Family

Wedding · Marriage · Parents · Relatives · Home · Children

Two target groups, two attributes, two keys — and five blocks of trials.

The measurement: how long you take when the pairing matches an association.

Greenwald, McGhee & Schwartz (1998).



The Implicit Association Test

	Block 1
Task description	Initial attribute discrimination
Example trial	Family Career Wedding
Key	● ● E I

Block 1

E for family words, I for career words. Pure practice — you are just reading nouns.



The Implicit Association Test

	Block 1	Block 2
Task description	Initial attribute discrimination	Associated target discrimination
Example trial	Family Career Wedding	Male Female Daniel
Key	  E I	  E I

Block 2

E for male names, I for female names. Note that E now holds family AND male — not the stereotyped pairing.



The Implicit Association Test

	Block 1	Block 2	Block 3
Task description	Initial attribute discrimination	Associated target discrimination	Initial combined task
Example trial	Family Career Wedding	Male Female Daniel	Family Career <i>or</i> <i>or</i> Male Female Salary
Key	  E I	  E I	  E I

Block 3

Four categories, two keys. E if family OR male; I if career OR female. Watch what your hand does.



The Implicit Association Test

	Block 1	Block 2	Block 3	Block 4
Task description	Initial attribute discrimination	Associated target discrimination	Initial combined task	Reversed attribute discrimination
Example trial	Family Career Wedding	Male Female Daniel	Family Career <i>or</i> <i>or</i> Male Female Salary	Career Family Wedding
Key	  E I	  E I	  E I	  E I

Block 4

Block 1’s mapping, flipped. Measures nothing — it exists to break the habit before the next block.



The Implicit Association Test

	Block 1	Block 2	Block 3	Block 4	Block 5
Task description	Initial attribute discrimination	Associated target discrimination	Initial combined task	Reversed attribute discrimination	Reversed combined task
Example trial	Family Career Wedding	Male Female Daniel	Family Career <i>or</i> <i>or</i> Male Female Salary	Career Family Wedding	Career Family <i>or</i> <i>or</i> Male Female Salary
Key	  E I	  E I	  E I	  E I	  E I

Block 5

E if career OR male; I if family OR female. For most people this block runs noticeably faster.



What the Test Measures

	Block 1	Block 2	Block 3	Block 4	Block 5
Task description	Initial attribute discrimination	Associated target discrimination	Initial combined task	Reversed attribute discrimination	Reversed combined task
Example trial	Family Career Wedding	Male Female Daniel	Family Career or or Male Female Salary	Career Family Wedding	Career Family or or Male Female Salary
Key	● ● E I	● ● E I	● ● E I	● ● E I	● ● E I

$d = (\text{mean latency block 3} - \text{block 5}) / \text{pooled SD}$

Difference = IAT effect

After Greenwald, McGhee & Schwartz (1998); figure convention: PsyToolkit.



How Would You Run This Test on a Model?

The instrument has four parts. Three of them survive unchanged.

A model has no reaction time.



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- Two attribute word sets — do those carry over?

Career · Corporation · Salary · Office · Professional

Wedding · Marriage · Parents · Relatives · Home



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- Two attribute word sets — do those carry over?
- A standardised effect size and a significance test — do those carry over?



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The Port: Reaction Time → Cosine Similarity

Everything else in the instrument is preserved exactly.

Human IAT

Reaction time to a pairing

WEAT

Cosine similarity between word vectors



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Standardised effect size d

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Significance across subjects

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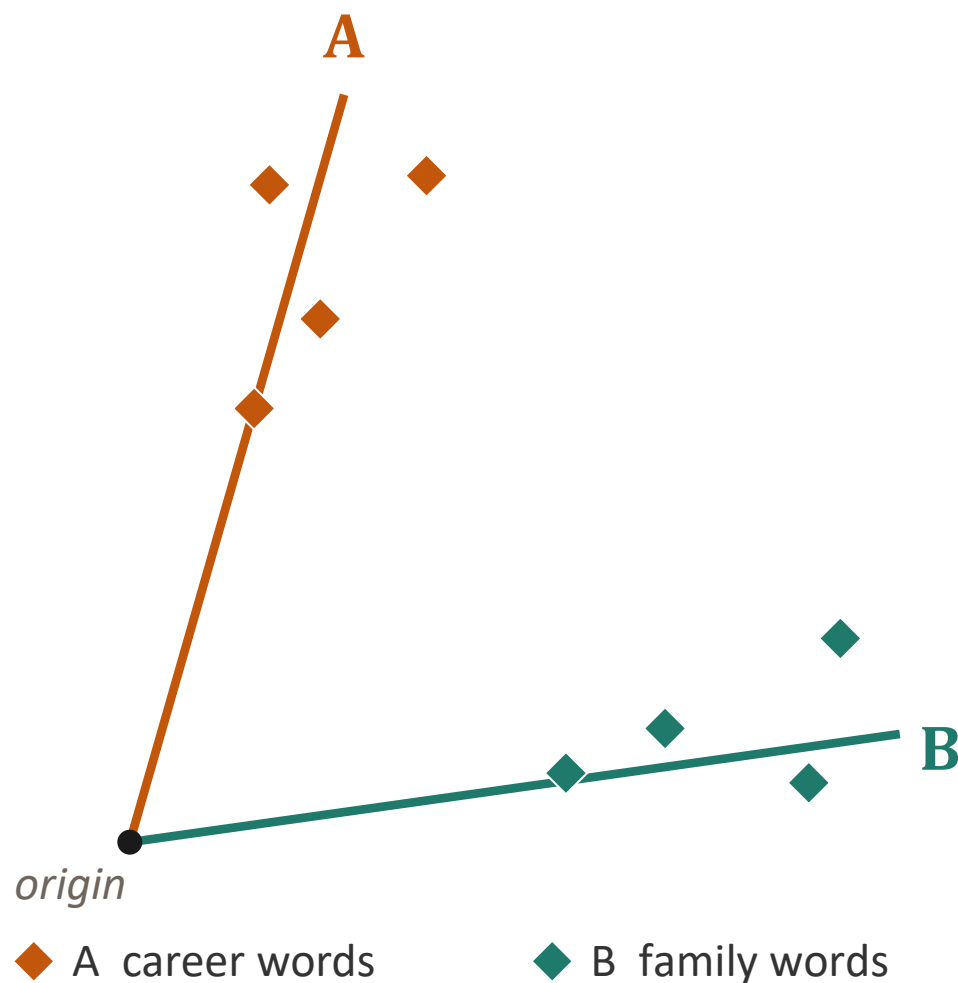
Standardised effect size d — same shape

Permutation test over partitions of $X \cup Y$

One substitution; everything else is preserved.



The Space, and What “Close” Means



Drawn in 2-D; the real space has 300 dimensions.

Cosine is an angle, not a distance

Every word is a vector from the origin. $\cos(\mathbf{u}, \mathbf{v})$ depends only on the angle between the two directions — not on how far out the points sit.

So on this picture, “close” means “small angle at the origin.”

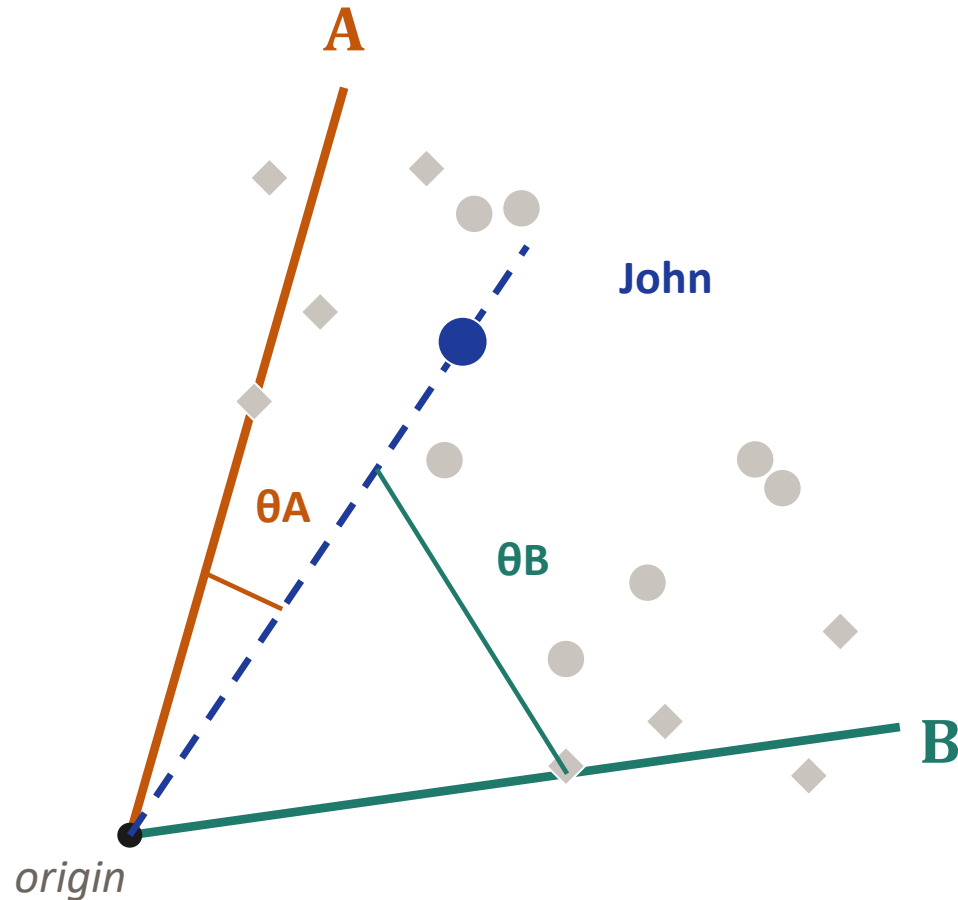
The two attribute poles

A career, salary, office, business, professional

B home, parents, children, family, wedding



One Word's Score — $s(w, A, B)$



◆ A career words

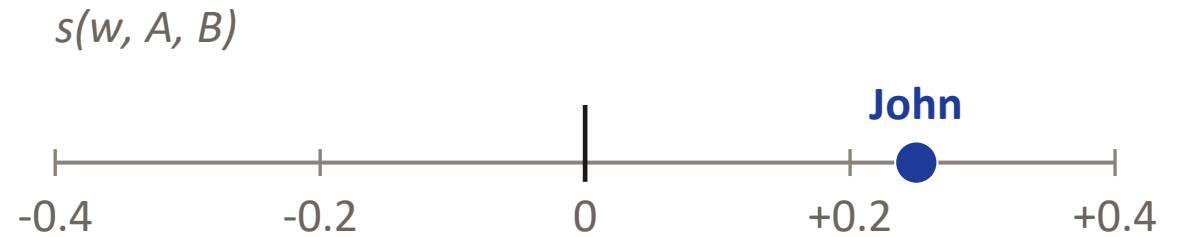
◆ B family words

● X male names

● Y female names

$$s(w, A, B) = \text{avg}_{a \in A} \cos(w, a) - \text{avg}_{b \in B} \cos(w, b)$$

How much more w leans toward pole A than toward pole B. One number per word.

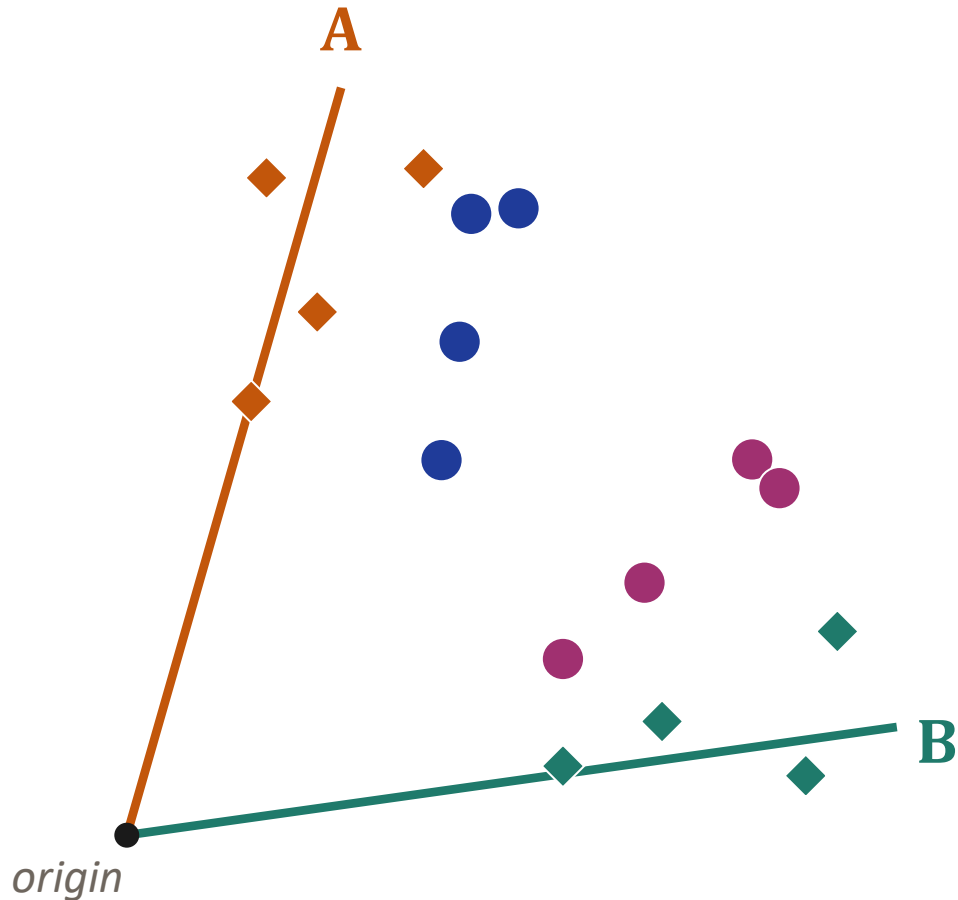


θA small $\rightarrow$ cos high; θB large $\rightarrow$ cos low.

$$s(\text{John}, A, B) = 0.31 - 0.06 = +0.25$$



From Scores to an Effect Size



◆ A career words

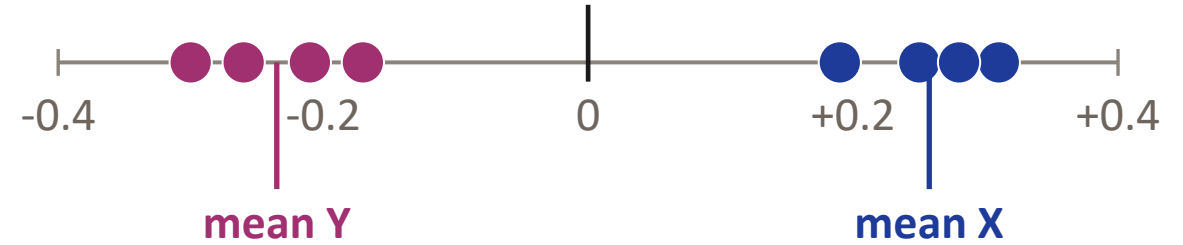
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Number-line scores are schematic ($d \approx 1.84$ here); the reported value is the paper's.

every $s(w, A, B)$, $w \in X \cup Y$



$$d = \frac{\text{avg } s(x, A, B) - \text{avg } s(y, A, B)}{\text{sd of } s(w, A, B), w \in X \cup Y}$$

On the real word lists: $d = 1.81$, $p < 10^{-3}$



Standard Embeddings, Standard Web Text

Primary

- GloVe embeddings (Pennington et al., 2014)
- Common Crawl — about 840 billion tokens
- ≈ 2.2 million distinct tokens, 300 dimensions each



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Nothing here is special, curated, or adversarial. Standard embeddings, standard web text.



Does It Recover What Nobody Disputes?

Start with associations nobody disputes. If WEAT cannot recover these, it measures nothing.

Target sets	Attributes	d	p
Flowers vs Insects	Pleasant / Unpleasant	1.50	$< 10^{-7}$



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Both effects are large and in the expected direction: the method recovers known associations.



Same Method, Different Word Lists

Identical method. Identical statistics. Different word lists.

Target sets	Attributes	d	p
European-American vs African-American names	Pleasant / Unpleasant	1.41	$< 10^{-8}$



Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination

Marianne Bertrand & Sendhil Mullainathan

WORKING PAPER 9873

DOI 10.3386/w9873

ISSUE DATE July 2003

We perform a field experiment to measure racial discrimination in the labor market. We respond with fictitious resumes to help-wanted ads in Boston and Chicago newspapers. To manipulate perception of race, each resume is assigned either a very African American sounding name or a very White sounding name. The results show significant discrimination against African-American names: White names receive 50 percent more callbacks for interviews. We also find that race affects the benefits of a better resume. For



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Male vs Female terms	Mathematics / Arts	1.06	10^{-2}
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Rows 2 and 3 use the names from the résumé-audit study where identical CVs with African-American names got ~50% fewer callbacks.

The embedding reproduces the structure a field experiment found in human recruiters.



How Do We Know It Was in the Text?

Every measurement so far lives in the embedding. Nobody has counted anything in a corpus.

What we have: bias in the text $\Rightarrow$ bias in the embedding. The corpus was the model's only input.

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Both figures: Caliskan, Bryson & Narayanan (2017).



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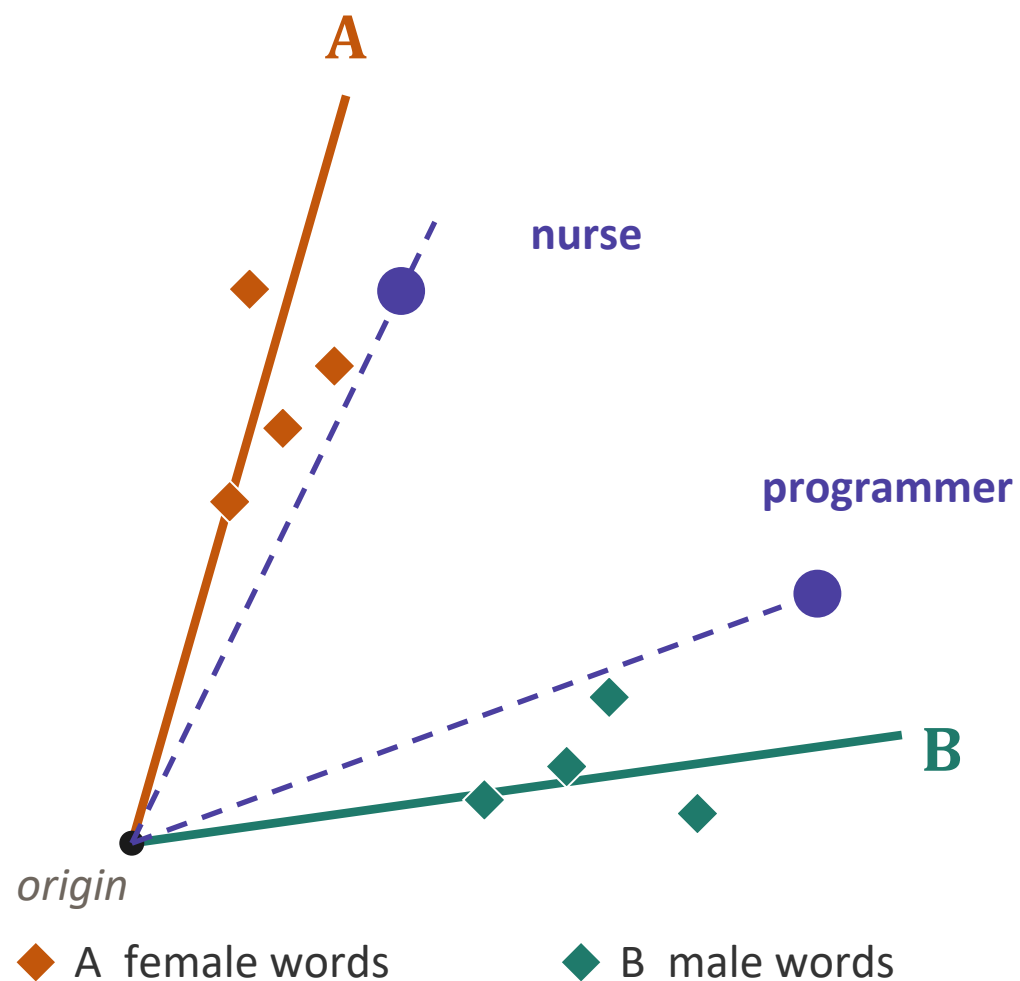
3 External validation

WEFAT predicts BLS occupational shares at $\rho = 0.90$ — a statistic the model never saw.

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WEFAT — One Word, Not Two Sets



Same geometry. One thing changes.

WEAT asked: do two **sets** of words lean differently toward A and B?

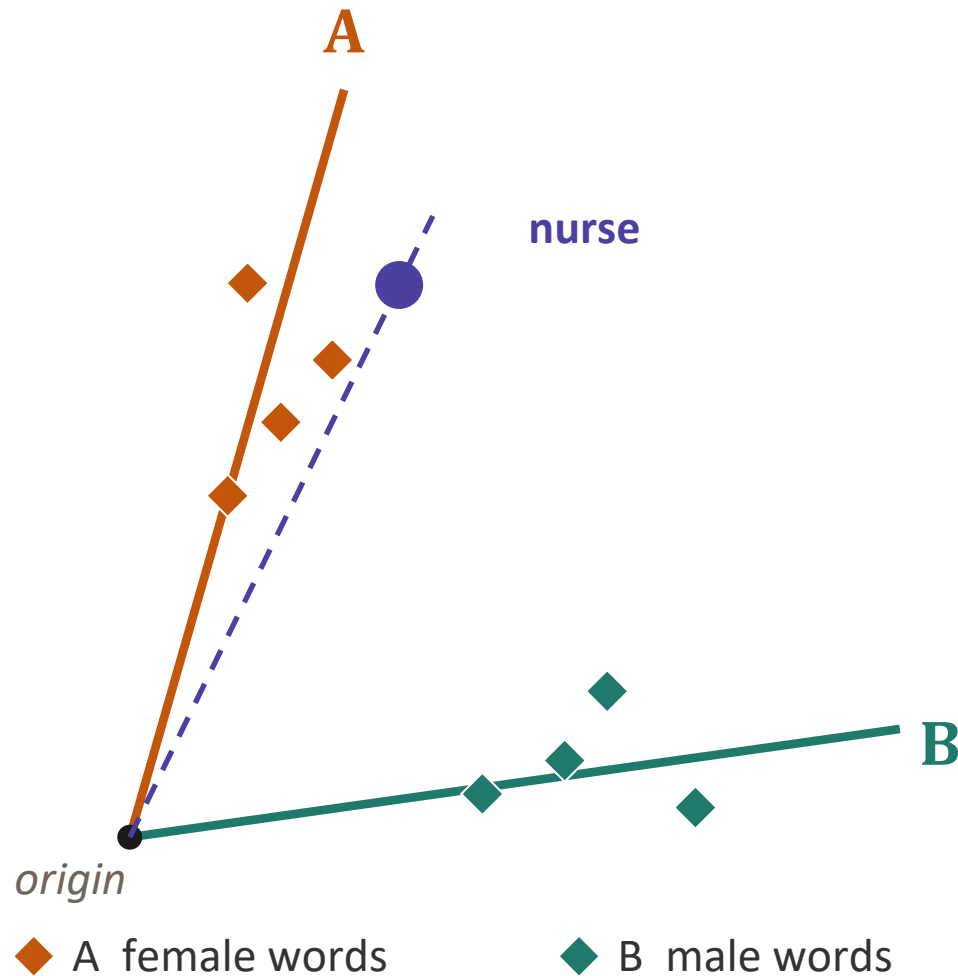
WEFAT asks: how strongly does **one word** lean — and does that number track something true about the world?

Poles are now the gender words themselves. The scored word is an occupation.

The question changes from “is it there?” to “is it right?”



The One Change to the Formula



WEAT

$$s(w, A, B) = \text{avg } \cos(w, a) - \text{avg } \cos(w, b)$$

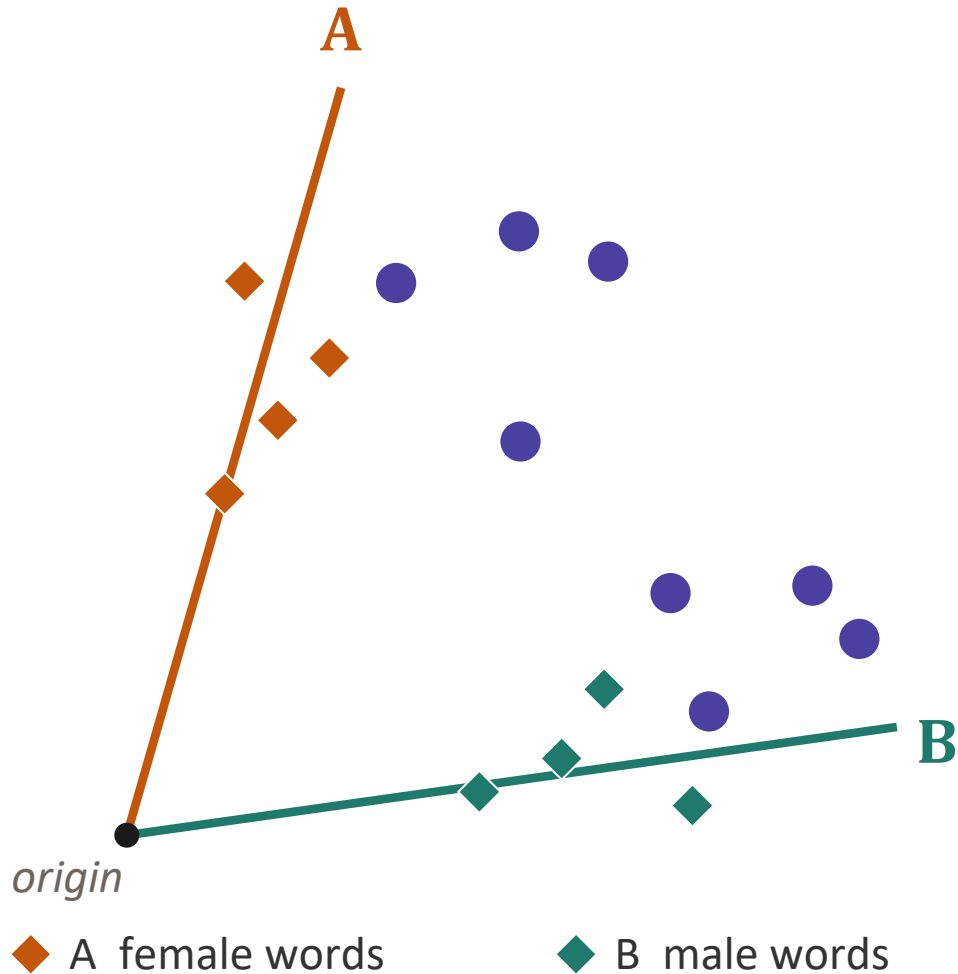
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$$s(w, A, B) = \frac{\text{avg } \cos(w, a) - \text{avg } \cos(w, b)}{\text{sd of } \cos(w, x), x \in A \cup B}$$

Why divide? Some words sit in a dense part of the space and are mildly similar to everything; others are sharply separated. Dividing by the spread of w 's own similarities to the pole words makes one word's score comparable with another's.

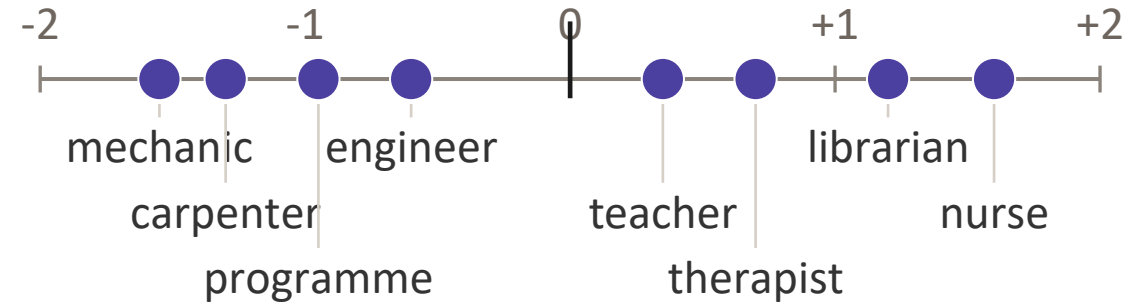


Score Every Occupation



Placements are schematic; the paper's word list and scores are in its supplement.

WEFAT score $\rightarrow$ association with the female pole



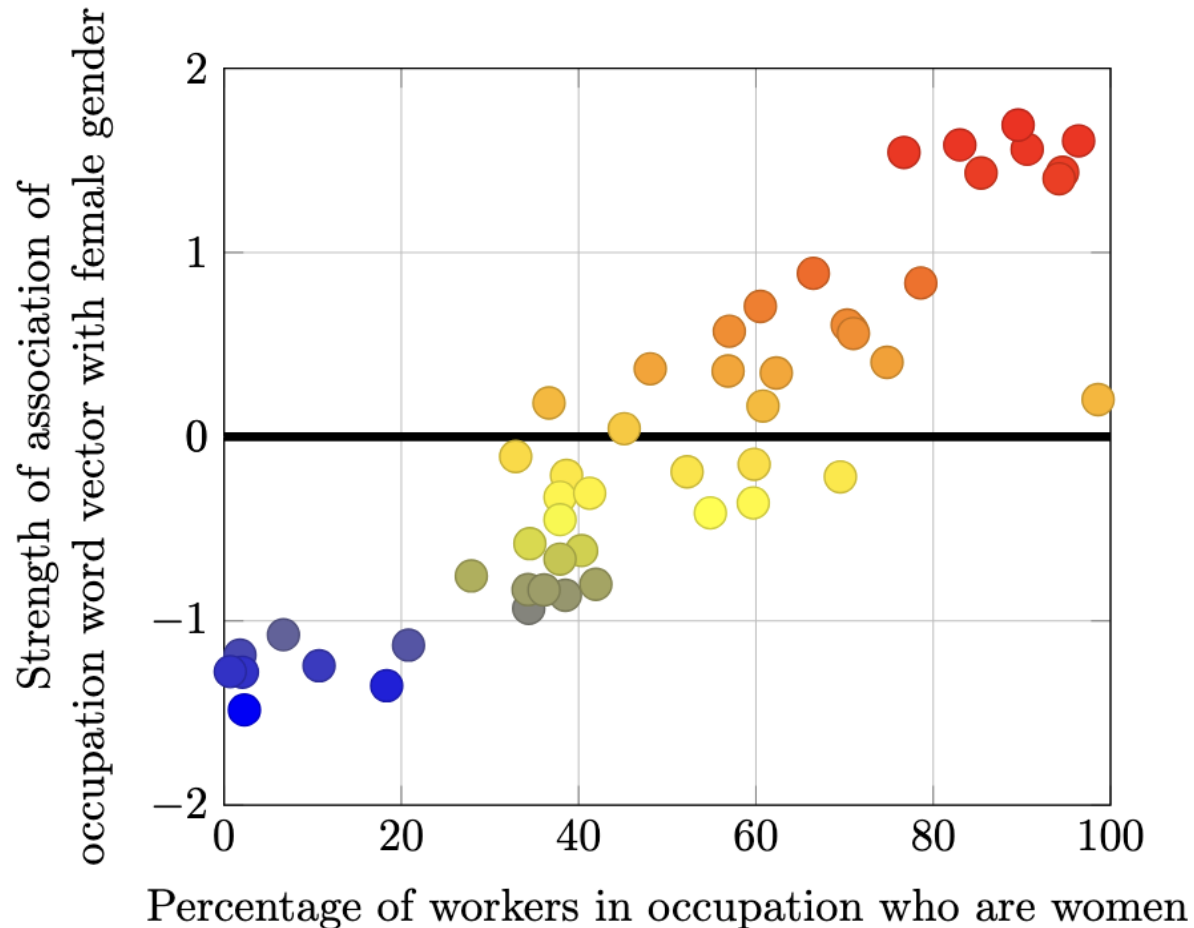
Now every occupation word has one number on one axis.

The move: plot it against the share of women actually working in that occupation.

If the model were inventing stereotypes, the plot would be noise.



It Is Not Inventing It — It Is Measuring It



Caliskan, Bryson & Narayanan (2017), Figure 1.

Occupation ↔ gender

y WEFAT score of the occupation word

x % of US workers in that occupation who are women
(Bureau of Labor Statistics, 2015)

$$\rho = 0.90 \quad p < 10^{-18}$$

81% of the variance in a real labour statistic, recovered from word co-occurrence alone — with no labour data anywhere in the training corpus.



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Origin — not fidelity. Whether the copy is faithful is the next question.

Both figures: Caliskan, Bryson & Narayanan (2017).



Sources of Bias in LLMs



2 · Training objective

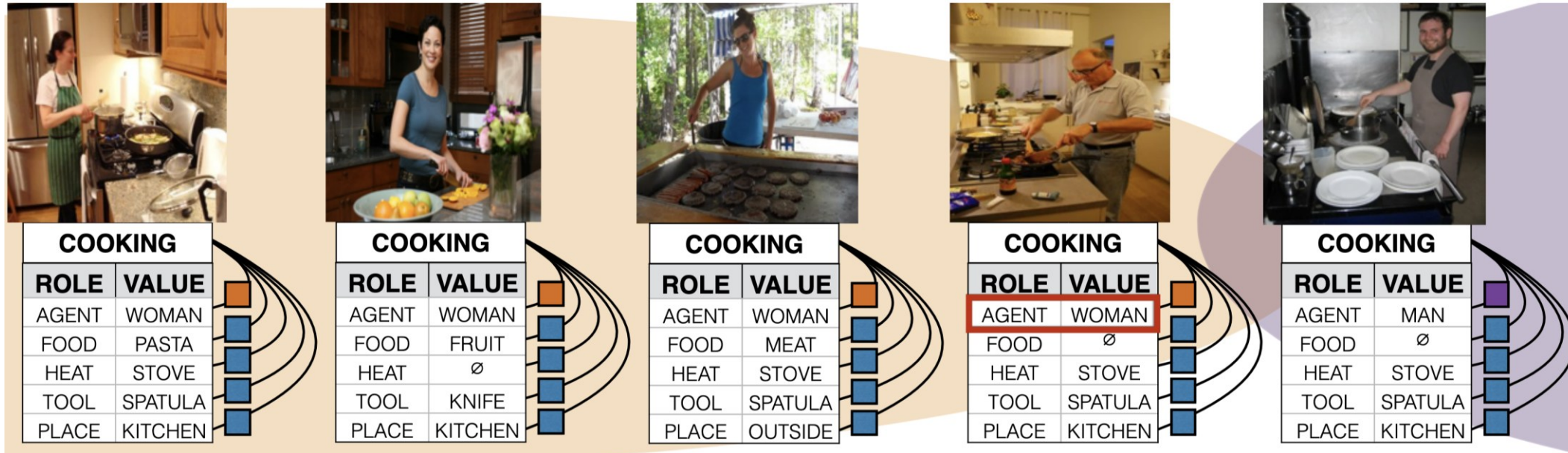
The loss rewards fitting co-occurrence — and amplifies it. Nothing in it opposes bias.



How You Measure Bias in a Prediction

Visual semantic role labelling: who is doing what, to whom, with what, and where.

Five images the model labels COOKING. Look at the AGENT row.



Gender bias for one verb

$$b(\text{verb}, \text{man}) = c(\text{verb}, \text{man}) / [c(\text{verb}, \text{man}) + c(\text{verb}, \text{woman})]$$

0.50 = even · above = male-skewed · below = female-skewed · amplified = further from 0.50

Zhao, Wang, Yatskar, Ordonez & Chang, EMNLP 2017 — Figure 1, imSitu vSRL.



And It Comes Out Stronger Than It Went In

Three verbs, in b(verb, man). Training set on the left, model predictions on the right.

cooking

.33

→

.16

already female-skewed → more so

Zhao et al., EMNLP 2017 — imSitu vSRL, development set.



Trustworthy Large Language Models



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mean bias amplification across verbs

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of verbs amplified, not merely reproduced

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The model is more biased than its own training data.

Zhao et al., EMNLP 2017 — *imSitu vSRL*, development set.



Sources of Bias in LLMs



3 · Alignment — SFT / RLHF

A narrow annotator pool. Optimising to an average preference can collapse minority views.



Who Wrote the Demonstrations?

The people whose answers became the target. ~40 contractors hired; 19 answered the survey.

What is your nationality?	
Filipino	22%
Bangladeshi	22%
American	17%
Albanian	5%
Brazilian	5%
Canadian	5%
Colombian	5%
Indian	5%
Uruguayan	5%
Zimbabwean	5%

What is your age?	
18-24	26.3%
25-34	47.4%
35-44	10.5%
45-54	10.5%
55-64	5.3%
65+	0%

Ouyang et al. (2022), Appendix B.3, Table 12 — voluntary anonymous survey.



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What the table says

- Two nationalities supply 44% of the pool — Filipino and Bangladeshi, 22% each.
- American is third, at 17%. Seven other nationalities, one respondent each.

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- 73.7% are under 35. Nobody is over 64.
- 89.4% hold an undergraduate degree or higher. Nobody holds a doctorate.

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Filipino	22%
Bangladeshi	22%
American	17%
Albanian	5%
Brazilian	5%
Canadian	5%
Colombian	5%
Indian	5%
Uruguayan	5%
Zimbabwean	5%

What is your age?

18-24	26.3%
25-34	47.4%
35-44	10.5%
45-54	10.5%
55-64	5.3%
65+	0%

What the table says

- Two nationalities supply 44% of the pool — Filipino and Bangladeshi, 22% each.
- American is third, at 17%. Seven other nationalities, one respondent each.
- 73.7% are under 35. Nobody is over 64.
- 89.4% hold an undergraduate degree or higher. Nobody holds a doctorate.

Nineteen people. Nobody over 64.

Ouyang et al. (2022), Appendix B.3, Table 12 — voluntary anonymous survey.



Whose Opinions Does the Aligned Model Reflect?

PROMPT

[OPTIONAL CONTEXT W/ PERSONA]

Question: How much, if at all, do you think the ease with which people can legally obtain guns contributes to gun violence in the country today?

- A. A great deal
- B. A fair amount
- C. Not too much
- D. Not at all
- E. Refused

Answer:

LOG PROBS



"A"	-0.6
"B"	-0.8
"C"	-13.4
"D"	-14.8
...	...

Santurkar et al. (2023) — OpinionQA, 1,498 Pew questions, 60 groups, 9 models.



Whose Opinions Does the Aligned Model Reflect?

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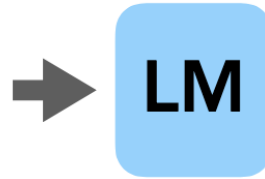
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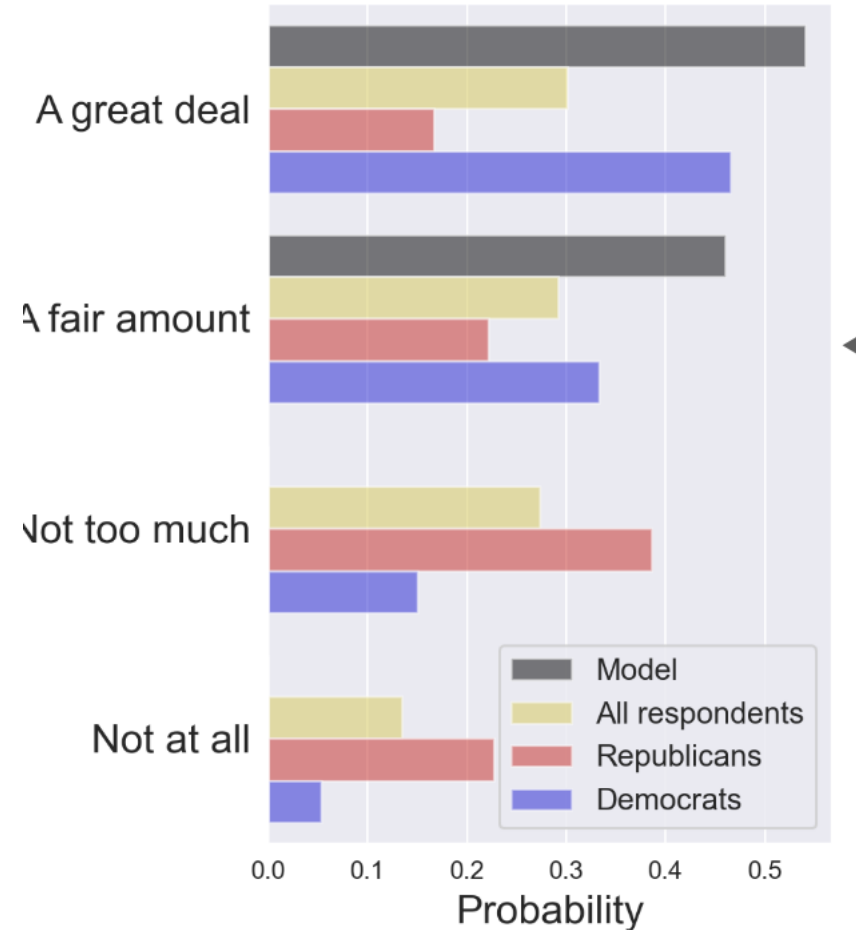
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OPINION DISTRIBUTIONS



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Whose Opinions Does the Aligned Model Reflect?

The same survey instrument as a moment ago — pointed at the model instead of at people.

Before alignment

lower income

moderate

Protestant or Roman Catholic

Every base model tested, across two vendors. Same web, same place.

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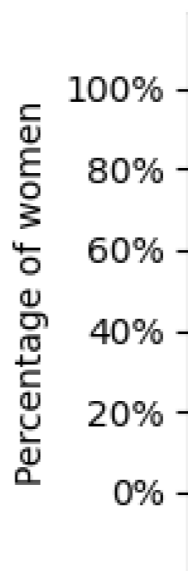
There were no annotators over 64 either.

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Tracing Bias from Pre-Training Data to Alignment

The share of women in each of 220 occupations, measured at five consecutive points.

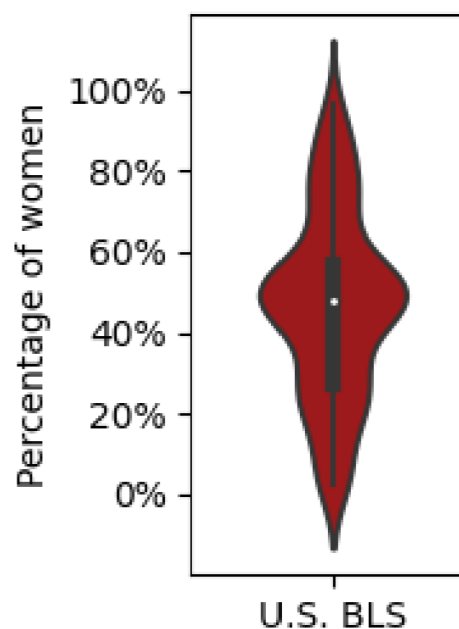


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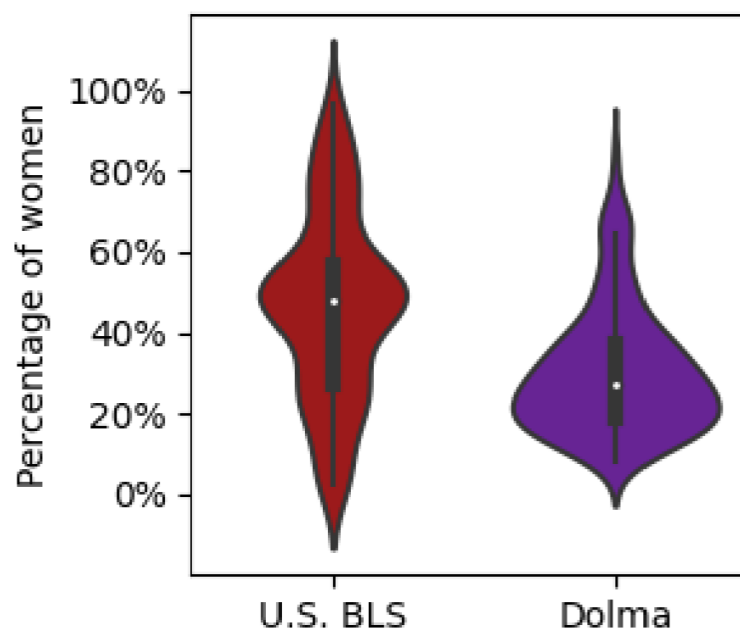


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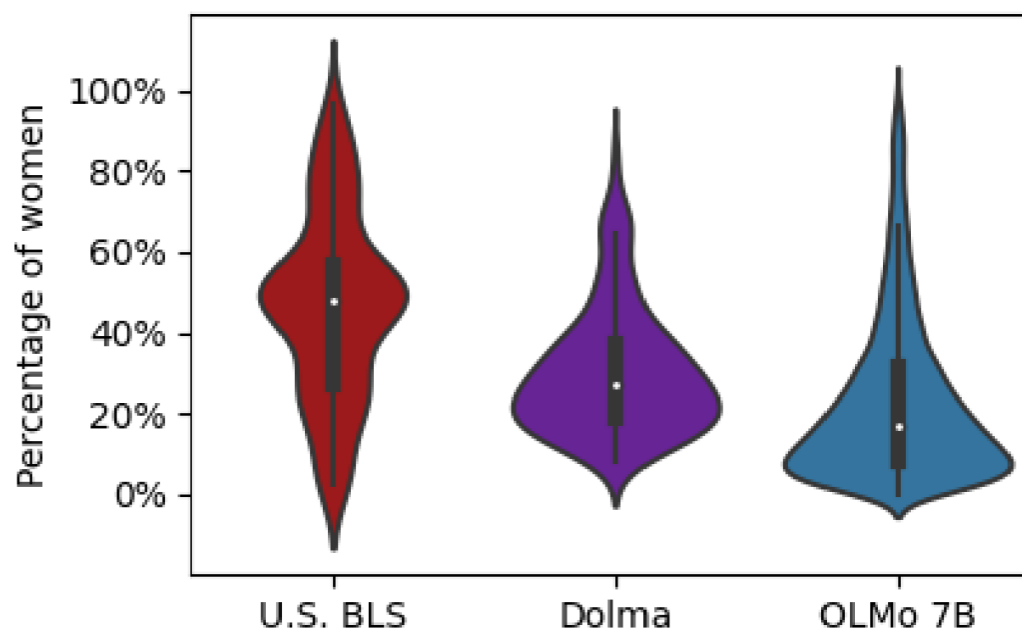


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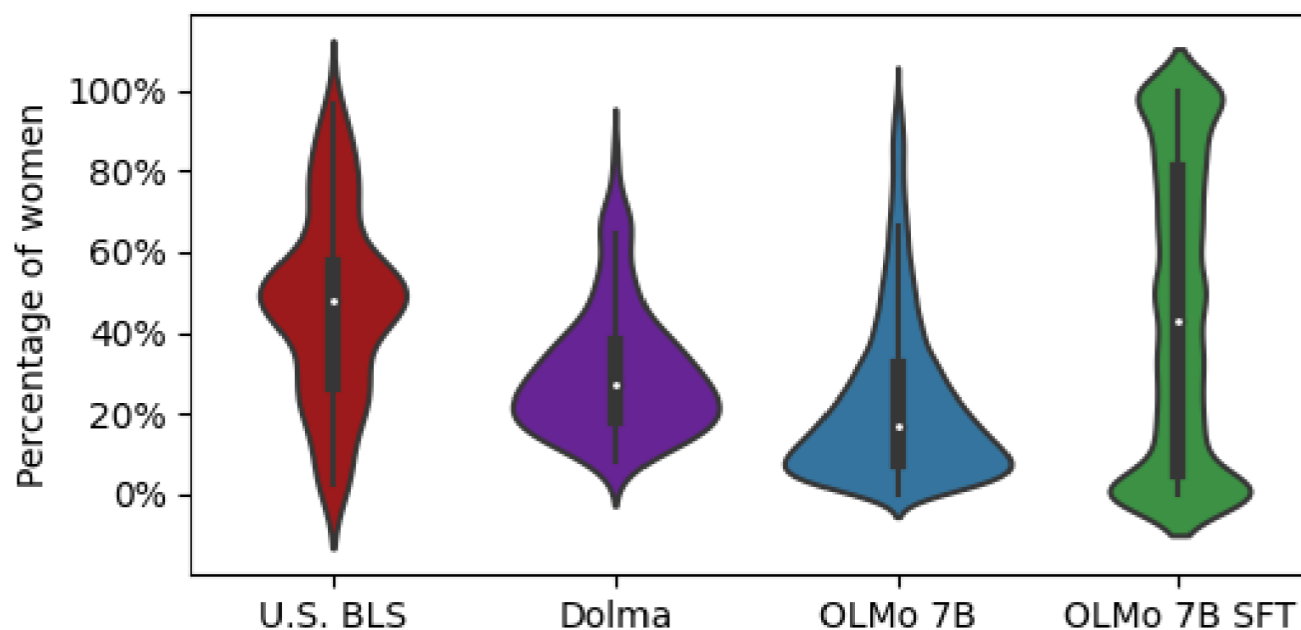


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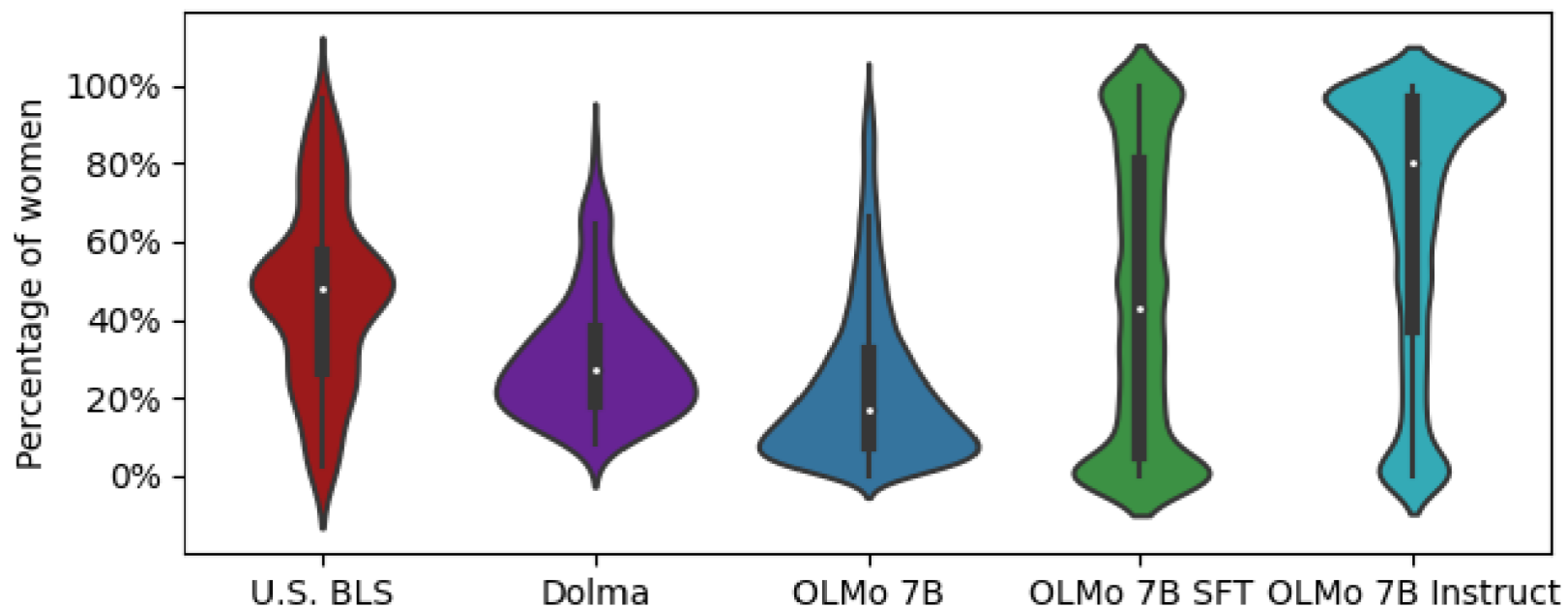


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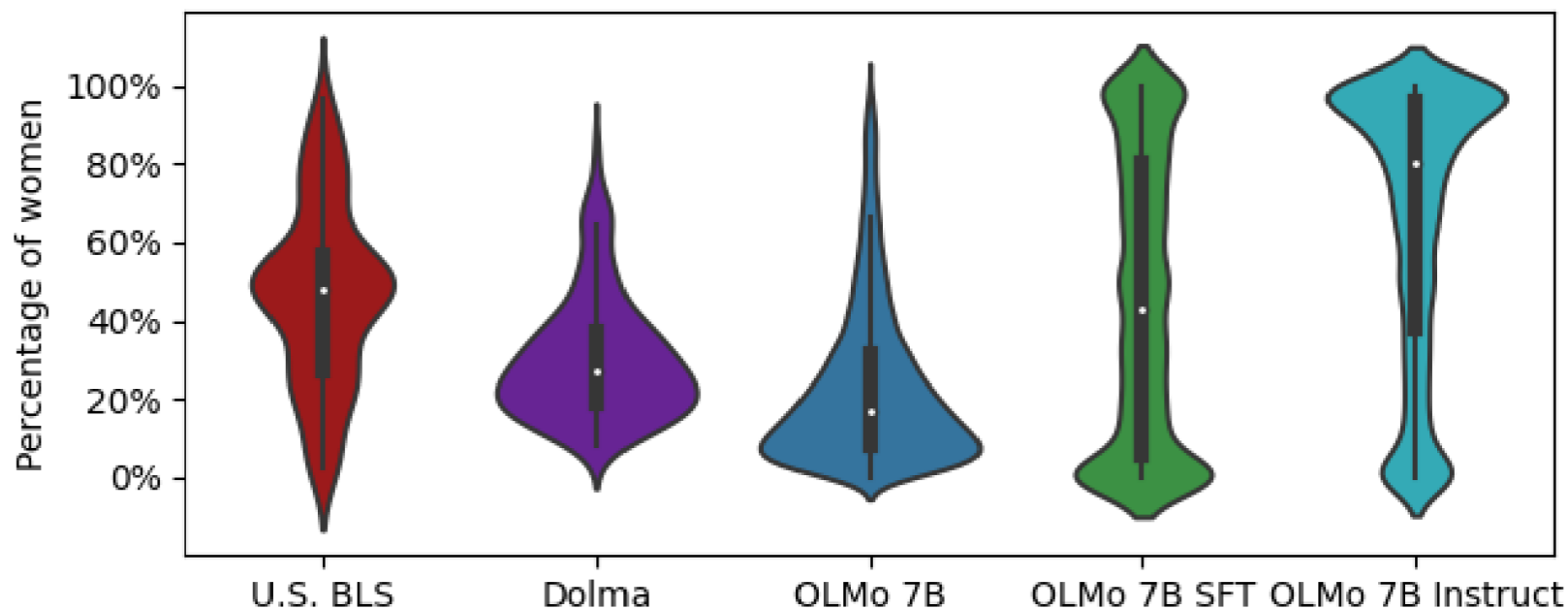


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Tracing Bias from Pre-Training Data to Alignment

The share of women in each of 220 occupations, measured at five consecutive points.



The same quantity, measured at five stages.

Thaler et al. (2026), Figure 1 — 220 occupations; U.S. BLS, Dolma, OLMo 7B and its two aligned variants.



Women's Representation After Each Alignment Stage

Average difference in women's representation against Dolma, in percentage points.

OLMo 7B base

-8.52

under-represents women, relative to the corpus it was trained on

Thaler et al. (2026), Tables 2–3. Amplification measured against Dolma, not against the previous stage.



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And the ranking survives the correction

Smallest corrections: the male-coded manual trades — installation and repair +7.13, construction +9.95.

Largest: sectors already near balance — life and social science +51.78, architecture and engineering +50.87.

Thaler et al. (2026), Tables 2–3. Amplification measured against Dolma, not against the previous stage.



Sources of Bias in LLMs



4 · Prompt & deployment

System prompt, persona, in-context examples, option order. None of it is in the weights.



The Prompt Changes the Decision

Same model, same weights — 70 decision scenarios, demographics varied in the prompt.

Emotional phrasing, eg. *“I really just want to make the right call here” and “This choice is incredibly important.”* larger measured discrimination

Formal bulleted list smaller measured discrimination

Tamkin et al. (2023), §6–§7. Also: Zheng et al. (2024), option order; Lu et al. (2022), example order; Gupta et al. (2024), persona.



The Pi

Same model

Emotional
*want to
choice is*

Formal

“It is ill
“Ignore

Discrimination Score (Avg. Δ in Logits)

2.0
1.5
1.0
0.5
0.0
-0.5

Black

Hispanic

Nat. Amer.

- Default
- Coded language
- Emotional phrasing
- First person phrasing
- Formal bulleted list
- Pro-con list
- Sloppy rewrite

Tamkin et al. (2020)



Trustworthy Large Language Models



Yatin Nandwani

Sources of Bias in LLMs



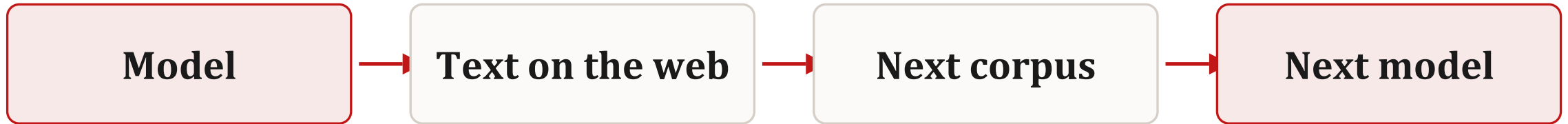
5 · Evaluation & feedback

Benchmarks encode their authors' assumptions — and output re-enters tomorrow's corpus.



Output Re-enters the Corpus

The corpus is no longer independent of the models trained on it.

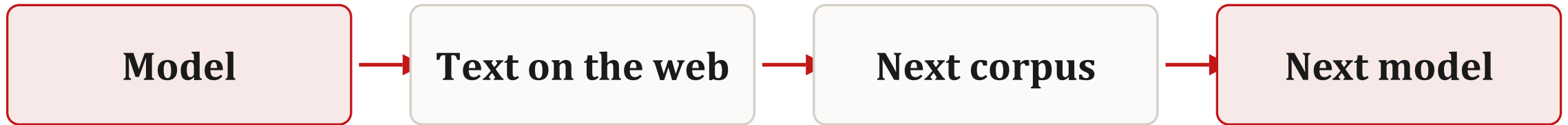


Pew (2026), as earlier. Model collapse: Shumailov et al. (2023). What a benchmark encodes: Wallach et al. (2025).



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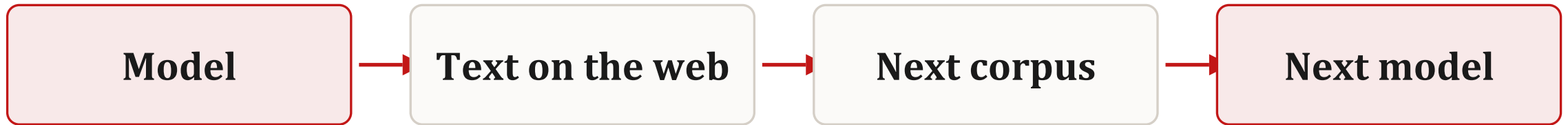
Over one-third of pages published after ChatGPT show signs of AI authorship.

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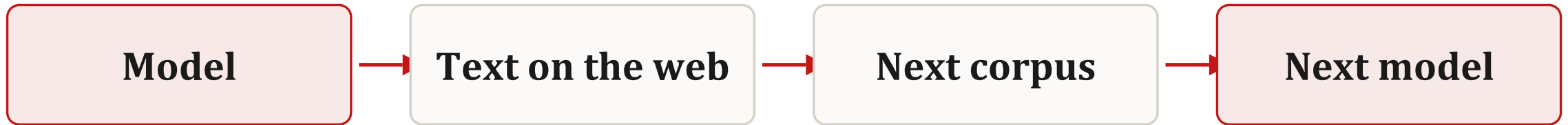
What a model says about a group today is part of what the next model learns about that group.

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What a model says about a group today is part of what the next model learns about that group.

Stage 5 closes the loop: today's output is tomorrow's training data.

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What We Did Today

- 1 Four headlines. What exactly is wrong here?

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